PHYSICS

1) A particle of mass m is under the influence of a force F which varies with the displacement x according to the relation $F = -kx + F_0$, in which k and F_0 are constants . The particle when disturbed from equilibrium will oscillate :-

(1) About x = 0, with $\omega \neq \sqrt{k/m}$

(2) About x = 0, with $\omega = \sqrt{k/m}$

(3) About $x = F_0/k$ with $\omega = \sqrt{k/m}$

(4) About $x = F_0/k$ with $\omega \neq \sqrt{k/m}$

2)

The phase difference between the two simple harmonic oscillations, $y_1 = \frac{1}{2}\sin \omega t + \left(\frac{\sqrt{3}}{2}\right)\cos \omega t$ and $y_2 = \sin \omega t + \cos \omega t$ is :-

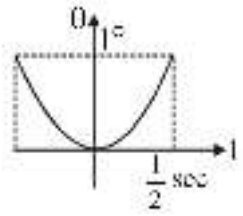
(1) $\frac{\pi}{6}$

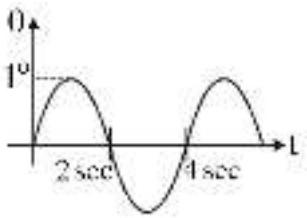
(2) $-\frac{\pi}{6}$

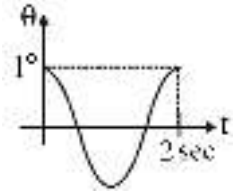
(3) $\frac{\pi}{12}$

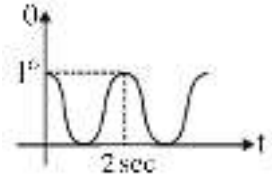
(4) $\frac{7\pi}{12}$

3) A simple pendulum has a string of length 1m. If the angular amplitude is 1°, which of the following can represent its angular displacement as a function of time :-

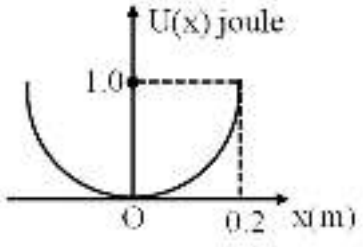
(1) 

(2) 

(3) 

(4) 

4) A particle of mass 4kg moves simple harmonically such that its PE (U) varies with position x, as shown. The period of oscillations is :-



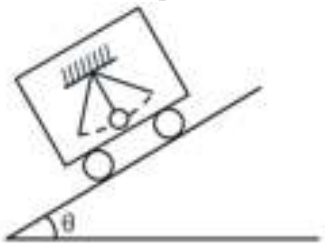
(1) $\frac{2\pi}{25}$ s

(2) $\frac{\pi\sqrt{2}}{5}$ s

(3) $\frac{4\pi}{5}$ s

(4) $\frac{2\sqrt{2}\pi}{5}$ s

5) Find time period of simple pendulum executing SHM as shown in figure :-



(1) $2\pi\sqrt{\frac{\ell}{g\sin\theta}}$

(2) $2\pi\sqrt{\frac{\ell}{g\cos\theta}}$

(3) $2\pi\sqrt{\frac{\ell}{g\sqrt{1+\sin^2\theta}}}$

(4) $2\pi\sqrt{\frac{\ell}{g(1+3\sin^2\theta)}}$

6) A simple pendulum oscillating in air has period T. The bob of the pendulum is completely

immersed in a non-viscous liquid. The density of the liquid is $\left(\frac{1}{16}\right)^{th}$ of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is :

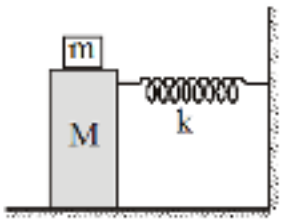
(1) $4T\sqrt{\frac{1}{15}}$

(2) $2T\sqrt{\frac{1}{10}}$

(3) $4T\sqrt{\frac{1}{14}}$

(4) $2T\sqrt{\frac{1}{14}}$

7) In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k. The mass oscillates on a frictionless surface with time period T and amplitude A. When the mass is in equilibrium position, (as shown in the figure) another mass m is gently fixed upon it. The new amplitude of oscillation will be :



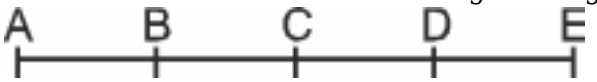
(1) $A\sqrt{\frac{M-m}{M}}$

(2) $A\sqrt{\frac{M}{M+m}}$

(3) $A\sqrt{\frac{M+m}{M}}$

(4) $A\sqrt{\frac{M}{M-m}}$

8) A body performs SHM along the straight line segment ABCDE with C as the mid point of segment AE (A and E are the extreme position for the SHM). Its kinetic energies at B and D are each one fourth of its maximum value. If length of segment AE is 2R, then the distance between B and D is :



(1) $\frac{\sqrt{3}}{2}R$

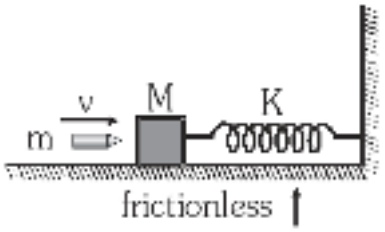
(2) $\frac{R}{\sqrt{2}}$

(3) $\sqrt{3}R$

(4) $\sqrt{2}R$

9) A blocks of mass M, at rest on a horizontal frictionless table, is attached to a rigid support by a spring of constant K. A bullet of mass m and velocity v strikes the block as shown in fig. The bullet

remains embedded in the block. Then the amplitude of the resulting simple harmonic motion is-



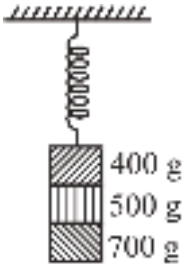
(1) $\frac{mv}{K(m+M)}$

(2) $\frac{Mv}{K(m+M)}$

(3) $\frac{mv}{\sqrt{K(m+M)}}$

(4) $\frac{Mv}{\sqrt{K(m+M)}}$

10) Three masses 700 gm, 500 gm and 400 gm are suspended at the end of a spring as shown and are in equilibrium. When the 700 gm mass is removed, the system oscillates with a period of 3 seconds; when the 500 gm mass is also removed, it will oscillate with a period of :-



(1) 1 sec

(2) 2 sec

(3) 3 sec

(4) $\sqrt{\frac{12}{5}}$ sec

11) **Assertion:** Damped vibrations indicate loss of energy.
Reason: The loss of energy may be due to friction, air resistance etc.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

12) In a resonance tube, the first resonance with a tuning fork occurs at 16 cm and second at 49 cm. If the velocity of sound is 330 m/s, the frequency of tuning fork is (in Hz) :-

(1) 500

(2) 300

(3) 330

(4) 165

13) A cylindrical tube open at both ends has a fundamental frequency 400 Hz in air. The tube is dipped vertically in water so that half of its length is in water. The fundamental frequency of air-column is now :-

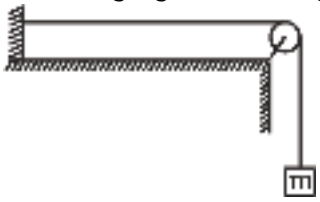
(1) 200 Hz

(2) 400 Hz

(3) 800 Hz

(4) 300 Hz

14) A transverse wave is passing through a string, as shown in figure. Mass density of the string is 1 kg/m^3 and cross section area of string is $0.01m^2$. Equation of wave in string is $y = 2\sin(20t - 10x)$. The hanging mass (in kg) is ['t' is in second and 'x' is in meter]:-



(1) 40

(2) 0.2

(3) 0.004

(4) 0.4

15) Velocity of sound waves in air is 330 m/sec. For a particular sound in air, a path difference of 40 cm is equivalent to a phase difference of 1.6π. The frequency of this wave is :-

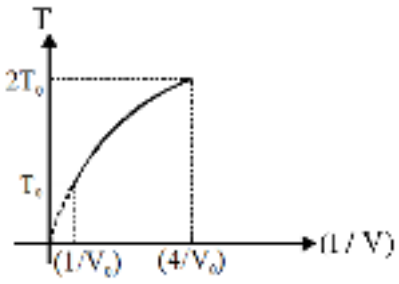
(1) 165 Hz

(2) 150 Hz

(3) 660 Hz

(4) 330 Hz

16) Figure shows a parabolic graph between T and $\frac{1}{V}$ for a mixture of a gas undergoing an adiabatic process. What is the ratio of v_{rms} and speed of sound in the mixture ?



(1) $\sqrt{\frac{3}{2}}$

(2) $\sqrt{2}$

(3) $\sqrt{\frac{2}{3}}$

(4) $\sqrt{3}$

17) Consider a Carnot cycle operating between T_1 = 500 K and T_2 = 300 K producing 1 kJ of mechanical work per cycle. Find the heat transferred to the engine by the reservoirs :-

(1) 1000 J

(2) 2000 J

(3) 1500 J

(4) 2500 J

18) A string is clamped at both the ends and it is vibrating in its 4th harmonic. The equation of the stationary wave is
Y = 0.3 sin(0.157x) cos(200πt).
The length of the string is : (All quantities are in SI units.)

(1) 20 m

(2) 80 m

(3) 60 m

(4) 40 m

19) When two waves of almost equal frequencies n_1 and n_2 are produced simultaneously, then the time interval between successive maxima is :-

(1) $\frac{1}{n_1 - n_2}$

(2) $\frac{1}{n_1} - \frac{1}{n_2}$

(3) $\frac{1}{n_1} + \frac{1}{n_2}$

(4) $\frac{1}{n_1 + n_2}$

20) In interference of two waves ; if $\frac{I_1}{I_2} = \frac{169}{100}$, then $\frac{I_{max}}{I_{min}}$ is :

(1) $\frac{23}{10}$

(2) $\frac{13}{10}$

(3) $\left(\frac{23}{3}\right)^2$

(4) $\frac{23}{7}$

21) A vibrating tuning fork of frequency n is placed near the open end of a long cylindrical tube. The

tube has a side opening and is also fitted with a movable reflecting piston. As the piston is moved through 8.75 cm, the intensity of sound changes from a maximum to minimum. If the speed of sound is 350 metre per second, then n is :-



(1) 500 Hz

(2) 1000 Hz

(3) 2000 Hz

(4) 4000 Hz

22) $y(x,t) = \frac{0.8}{(4x+5t)^2+5}$ represents a moving pulse, where x,y, are in metre and t in second. Then :

(i) pulse is moving in +x-direction
(ii) in 2 second, it travels a distance of 2.5 m
(iii) its maximum displacement is 0.16 m
(iv) it is a travelling wave pulse

(1) (i,iv)

(2) (ii,iii,iv)

(3) (i,ii,iv)

(4) (i,iii)

23) Equation of a plane progressive wave is given by

$y = 0.6\sin 2\pi\left(t-\frac{x}{2}\right)$. On reflection from a denser medium its amplitude becomes 2/3 of the amplitude of the incident wave. The equation of the reflected wave is -

(1) $y = 0.6 \sin 2\pi\left(t+\frac{x}{2}\right)$

(2) $y = -0.4 \sin 2\pi\left(t+\frac{x}{2}\right)$

(3) $y = 0.4 \sin 2\pi\left(t+\frac{x}{2}\right)$

(4) $y = -0.4 \sin 2\pi\left(t-\frac{x}{2}\right)$

24) A number of tuning forks are arranged in the order of increasing frequency and any two successive tuning forks produce 4 beats per second, when sounded together. If the last tuning fork has a frequency octave higher than that of the first tuning fork and the frequency of the first tuning fork is 256 Hz, then the number of tuning forks is :

(1) 63

(2) 64

(3) 65

(4) 66

25) Find ratio of velocity of sound in H_2 & O_2 at same temperature :-

(1) $\frac{1}{4}$

(2) $\frac{4}{1}$

(3) $\frac{1}{1}$

(4) $\frac{32}{1}$

26) The rates of cooling of a body at temperatures 100°C and 80°C are x_1 and x_2 respectively, when placed in a room of temperature 40°C, then $\frac{x_1}{x_2}$ is :-

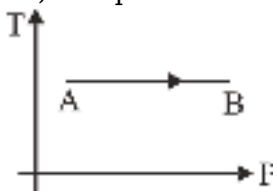
(1) 4/5

(2) 5/4

(3) 3/2

(4) 2/3

27) Temperature versus pressure graph of an ideal gas is shown in figure. During the process AB



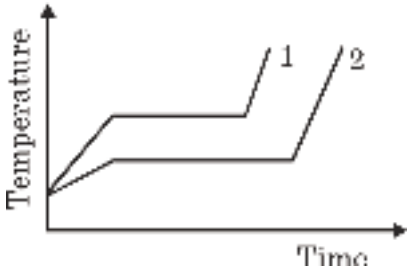
(1) Internal energy of the gas increases.

(2) Volume of the gas is increased

(3) Work done on the gas is positive

(4) Pressure is proportional to volume.

28) Two solid objects of the same mass are supplied with heat at the same rate ΔQ / Δt. The temperature of the first object with latent heat L_1 and specific heat capacity c_1 changes according to graph 1 on the diagram. The temperature of the second object with latent heat L_2 and specific heat capacity c_2 changes according to graph 2 on the diagram. Based on what is shown in the graph, the latent heats L_1 and L_2, and the specific heat capacities c_1 and c_2 in solid state obey which of the following relationships ?



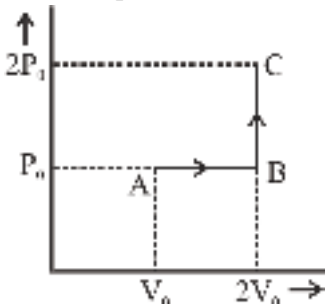
(1) $L_1 > L_2$; $c_1 < c_2$;

(2) $L_1 < L_2$; $c_1 < c_2$;

(3) $L_1 < L_2$; $c_1 > c_2$;

(4) $L_1 > L_2$; $c_1 > c_2$;

29) One mole of an ideal monoatomic gas is taken from A to C along the path ABC. The temperature of the gas at A is $2T_0$. For the process ABC (where R is gas constant), which of the following statement is wrong ?



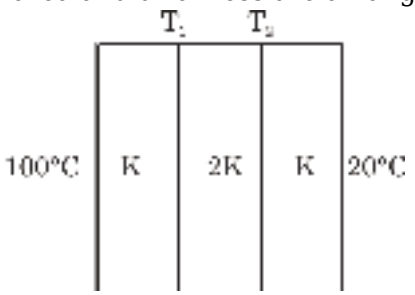
(1) Change in internal energy of gas is $9RT_0$

(2) Heat absorbed by the gas is $11RT_0$

(3) Work done by the gas is $2RT_0$

(4) Heat absorbed by the gas is $\frac{11}{12}RT_0$

30) Three slabs of equal area and thickness are arranged as shown in the figure. Find the value of T_1 and T_2 in steady state :-



(1) 68°C & 52°C

(2) 62°C & 58°C

(3) 60°C & 50°C

(4) 50°C & 30°C

31) The ratio of average translational K.E. to rotational K.E. of a linear polyatomic molecule at temperature T is :-

(1) 3

(2) 5

(3) $\frac{3}{2}$

(4) $\frac{7}{5}$

32) The power emitted per unit area of a black body is R watt/m^2. At what wavelength will the power radiated by the black body be maximum? If the Stefan's constant is σ and Wien's constant is b, then :-

(1) $\lambda = \left(\frac{b\sigma}{R}\right)^{1/4}$

(2) $\lambda = \left(\frac{b}{R}\right)^{1/4}$

(3) $\lambda = \left(\frac{bR}{\sigma}\right)^{1/4}$

(4) $\lambda = b\left(\frac{\sigma}{R}\right)^{1/4}$

33) The specific heat of a metal at low temperatures varies according to $S = (4/5)T^3$, where T is the absolute temperature. Find the heat energy (in SI unit) needed to raise temperature of unit mass of the metal from T = 1 K to T = 2 K.

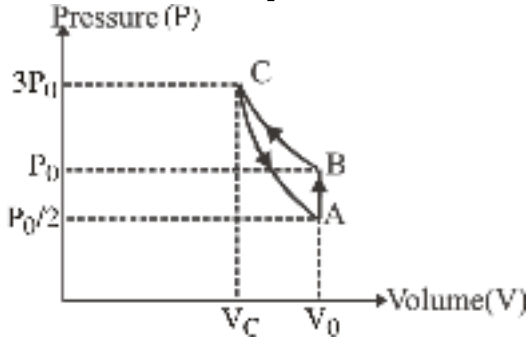
(1) 3J

(2) 4J

(3) 5J

(4) 6J

34) One mole of an ideal gas is carried through a thermodynamic cycle as shown in the figure. The cycle consists of an isochoric, an isothermal and an adiabatic processes. The adiabatic exponent of the gas is γ. Choose the correct option(s).



(1) $\gamma = \frac{\ln 6}{\ln 3}$

(2) $\gamma = \frac{\ln 5}{\ln 3}$

(3) BC is adiabatic

(4) AC is isothermal

35) The specific heat of an object of mass 2kg is 5 cal/kg °C. If an electric heater consumes 4200 J of electrical energy in 1 sec and converts 20% of energy to heat which is used to heat the object for 10 s, then the change in temperature of the object is -

(1) 100°C

(2) 200°C

(3) 300°C

(4) 400°C

36) A sample of O_2 is at a pressure of 1 atm when the volume is 100 ml and its temperature is 27°C. What will be the temperature of the gas (in °C) if the pressure becomes 2 atm and volume remains 100 ml.

(1) 600 °C

(2) 327 °C

(3) 400 °C

(4) 227 °C

37) A gas is filled in a cylinder, its temperature is increased by 20% on kelvin scale and volume is reduced by 10%. How much percentage of the gas will leak out at constant pressure ?

(1) 20%

(2) 25%

(3) 40%

(4) 50%

38) An ideal gas follows VT^2 = const. Its volume expansion coefficient will be :-

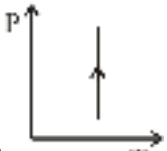
(1) $\frac{2}{T}$

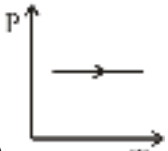
(2) $-\frac{2}{T}$

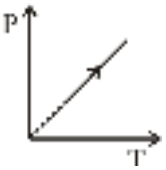
(3) $\frac{3}{T}$

(4) $-\frac{3}{T}$

39) Pressure versus temperature graphs of an ideal gas are as shown in figure. Choose the wrong statement:

(i) 

(ii) 

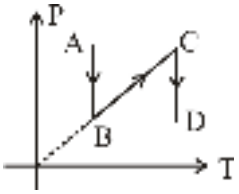
(iii) 

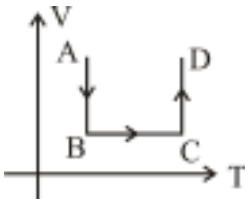
(1) Density of gas is increasing in graph (i)

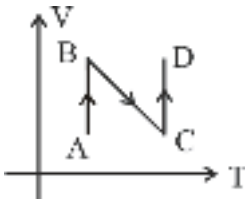
(2) Density of gas is decreasing in graph (ii)

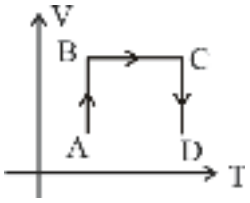
(3) Density of gas is constant in graph (iii)

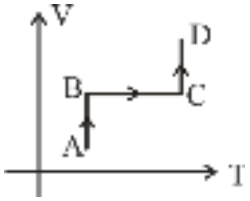
(4) None of these

40) P-T diagram is shown in given figure. Choose the corresponding V-T diagram 

(1) 

(2) 

(3) 

(4) 

41) A diatomic gas is supplied the heat Q very slowly keeping pressure constant. The work done by the gas will be -

(1) $\frac{2}{3}Q$

(2) $\frac{3}{5}Q$

(3) $\frac{2}{7}Q$

(4) $\frac{1}{5}Q$

42) For an ideal black body (IBB) power radiated at 0°C is P. What will be power radiated by black body at 546°C ?

(1) 16 P

(2) 81 P

(3) 27 P

(4) 64 P

43) Two liquids A and B are at 32°C and 24°C respectively. When equal masses of both are mixed then temperature of mixture becomes 28°C. The ratio of specific heat is :

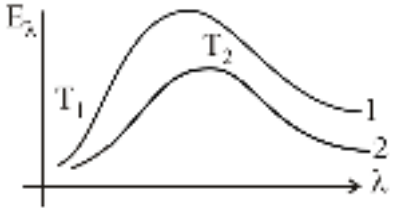
(1) 3 : 2

(2) 2 : 3

(3) 1 : 1

(4) 4 : 3

44) Spectral emissive power vs wavelength graph at two different temperature is as shown, then



(1) $T_1 > T_2$

(2) $T_1 < T_2$

(3) $T_1 = T_2$

(4) None

45) Length of a resonance tube is 130 cm. Speed of sound is 320 m/s and frequency of the source is 320 Hz.

	Column-I		**Column-II**
(a)	At resonance, minimum length of air column (in cm)	(i)	105
(b)	At resonance, minimum length of water column (in cm)	(ii)	3
(c)	At resonance, maximum length of water column (in cm)	(iii)	125
(d)	At resonance, maximum length of air column (in cm)	(iv)	25
(e)	Total number of resonance obtained is	(v)	5

Correct match between the columns is :

(1) a-(iv); b-(v); c-(i); d-(iii); e-(v)

(2) a-(iv); b-(i); c-(i); d-(iii); e-(ii)

(3) a-(iv); b-(v); c-(i); d-(iii); e-(ii)

(4) a-(iv); b-(iii); c-(i); d-(iv); e-(ii)

CHEMISTRY

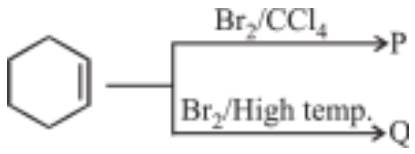
1) **Statement I :** Attack of electrophile on Benzene results in the formation of σ-complex in which one of the C is sp^3 hybridised.
Statement II : σ-complex is aromatic in nature.

(1) Statement I is correct and statement II is incorrect.

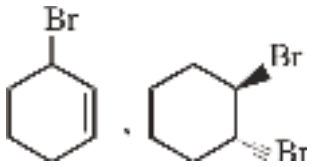
(2) Statement II is correct and statement I is incorrect.

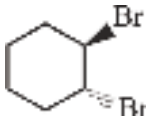
(3) Both statement are correct

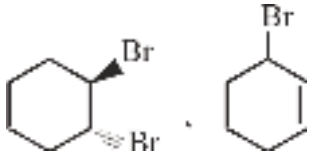
(4) Both statement are incorrect.

2) 

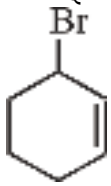
P and Q are respectively :-

(1) 

(2) P = Q = 

(3) 

P = Q =

(4) 

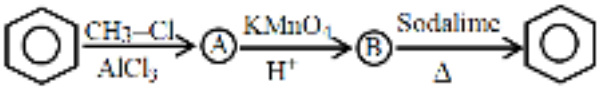
3) "Halogenation of methane take place at higher temperature or diffused sunlight". Find correct for this statement

(1) $F_2 < Cl_2 < Br_2 < I_2$ Reactivity of Halogen

(2) $CH_4 < CH_3 - CH_3 < CH_3 - CH_2 - CH_3$ Reactivity of H-atom

(3) Reaction initiate with heterolysis

(4) Intermediate is methyl carbocation.

4) 

Which of the following statement is incorrect ?

(1) A is Toluene

(2) Formation of A is Decarboxylation

(3) B is benzoic acid.

(4) Formation of A is Friedel Craft reaction

5) Match the following reactants in column-I with the corresponding reaction products in column-II and select the correct answer using the given codes below :

Column-I		**Column-II**	
(A)	Alkane (Excess) + $X_2 \xrightarrow{hv}$	(p)	water + carbondioxide
(B)	Alkane + $O_2 \xrightarrow{\Delta}$	(q)	Alkanal
(C)	Alkane + $O_2 \xrightarrow[\Delta]{Mo_2O_3}$	(r)	Haloalkane
(D)	n-hexane $\xrightarrow[10-20\ atm]{Cr_2O_3;773K}$	(s)	Benzene

Codes:

	A	**B**	**C**	**D**
(1)	p	q	r	s
(2)	p	q	s	r
(3)	r	p	q	s
(4)	r	p	s	q

(1) 1

(2) 2

(3) 3

(4) 4

6) Consider following table–

Column-I

(A) Best and latest technique for isolation, purification and separation of organic compounds

(B) In Lassaigne's test for 'N'

(C) In Lassaigne's test for 'S'

(D) Aniline is purified from water by

(E) Glycerol is purified from spent lye

Column-II

(i) Chromatography

(ii) Vacuum distillation

(iii) Violet coloured complex is formed

(iv) Prussian blue coloured complex is formed

(v) Steam distillation

The correct match is–

(A) (B) (C) (D) (E)

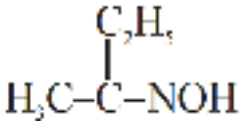
(1) (i) (ii) (iii) (iv) (v)
(2) (iv) (v) (iii) (ii) (i)
(3) (i) (iv) (iii) (v) (ii)
(4) (v) (iii) (iv) (ii) (i)

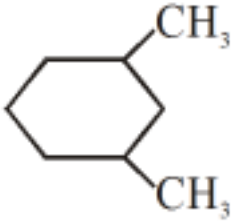
(1) 1
(2) 2
(3) 3
(4) 4

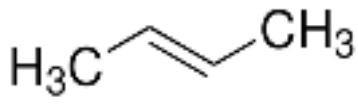
7) Which of the following reagents can not convert isobutane into isobutylene ?

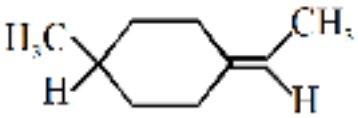
(1) Bromination in presence of ultra violet light followed by alc. KOH.
(2) Monochlorination followed by dehydrohalogenation
(3) Acidic $KMnO_4$ followed by conc. sulphuric acid in presence of heat.
(4) Hydroboration oxidation followed by dehydration with conc. sulphuric acid.

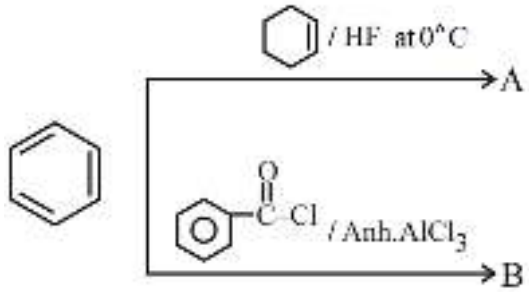
8) Which of following will not show geometrical isomerism ?

(1) 

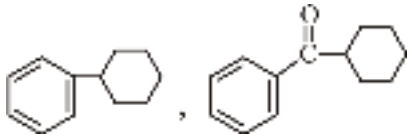
(2) 

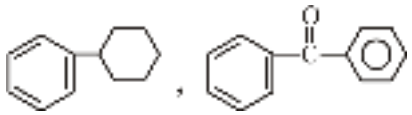
(3) 

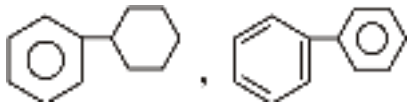
(4) 

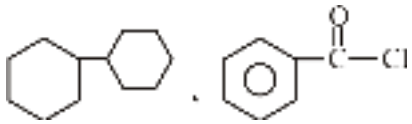
9) 

A & B respectively are :-

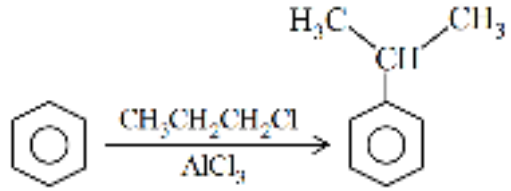
(1) 

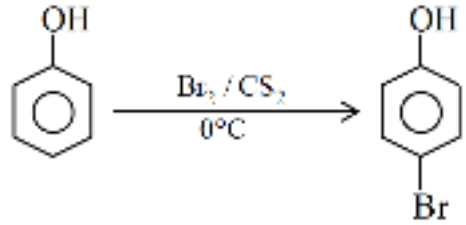
(2) 

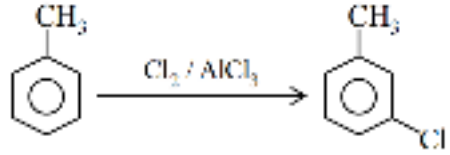
(3) 

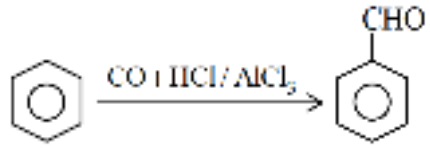
(4) 

10) In which of the following reaction, major product is wrong ?

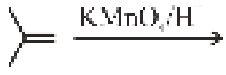
(1) 

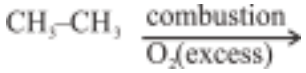
(2) 

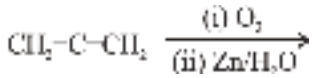
(3) 

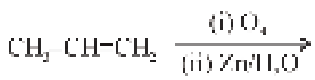
(4) 

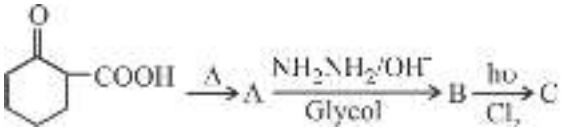
11) Which reaction does not produce CO_2 :-

(1) 

(2) 

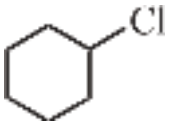
(3) 

(4) 

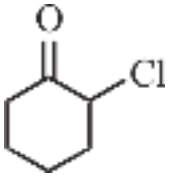
12) 

C is :-

(1) 

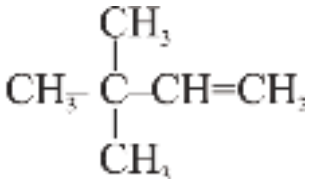
(2) 

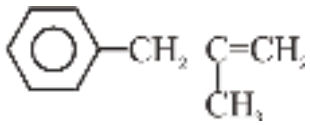
(3) 

(4) 

13) Which of the following alkene give 3° alcohol as major product on reaction with $H_3O^{\oplus}$:-

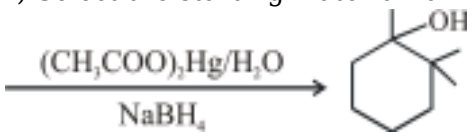
(1) CH_3–CH=CH–CH_3

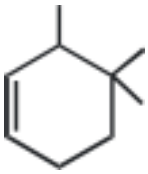
(2) 

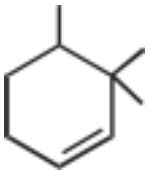
(3) 

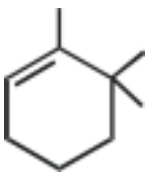
(4) Both (2) & (3)

14) Select the starting material for following reaction ?

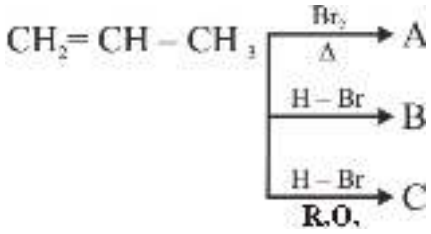
? 

(1) 

(2) 

(3) 

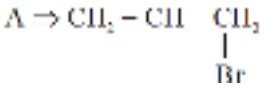
(4) Both (1) and (2)

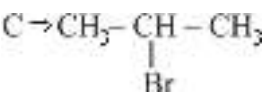
15) 

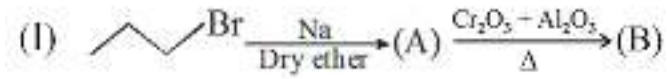
Which of the following statement is true?

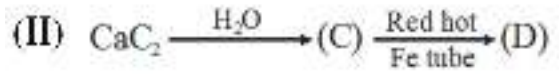
(1) A & B are position isomer of each other.

(2) B & C are identical

(3) 

(4) 

16) 



which of the following product is not formed in above reactions?

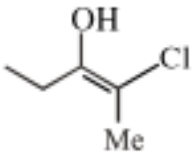
(1) 

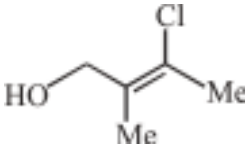
(2) 

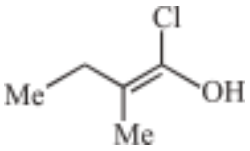
(3) CH ≡ CH

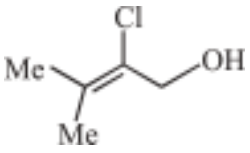
(4) 

17) Which structure represents
2-Chloro-3-methylbut-2-en-1-ol

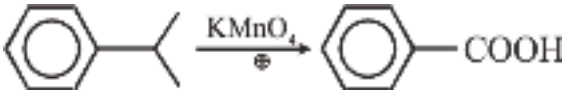
(1) 

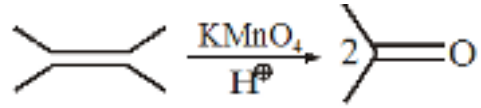
(2) 

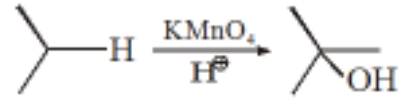
(3) 

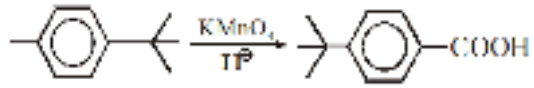
(4) 

18) Which of the following reaction is incorrect?

(1) 

(2) 

(3) 

(4) 

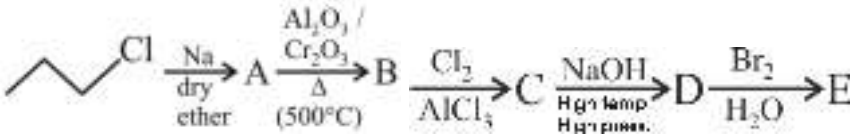
19) Which of the following statement is not correct in the mechanism of electrophilic substitution of benzene ?

(1) Benzene act as a Nucleophile

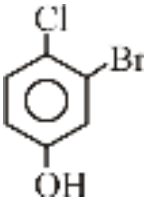
(2) Attack of electrophile resulting in formation of arenium ion in which one of the carbon is sp^3 hybridised

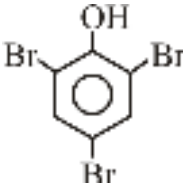
(3) Addition of hydride on benzene ring to give carbocation.

(4) Removal of proton from sp^3 carbon atom to restore aromatic character

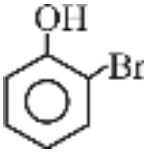
20) 

Final product will be ?

(1) 

(2) 

(3) 

(4) 

21) The Lassaigne's extract is boiled with conc HNO_3 while testing for the halogens. By doing so it :-

(1) decomposes Na_2S and NaCN, if formed

(2) helps in the precipitation of AgCl

(3) increases the solubility product of AgCl

(4) increases the concentration of NO_3^- ions

22) Which of the following compound can be separated by steam distillation method.:-

(1) Steam volatile but insoluble in water

(2) Steam volatile but soluble in water

(3) Steam non volatile but sparingly soluble in water

(4) Liquid in steam but solid in water

23) Technique for the purification of solid organic compounds based on difference in the solubilities of compound and the impurities in a suitable solvent.

(1) Sublimation

(2) Crystallisation

(3) Distillation

(4) Chromatography

24) Match the column :-

	Column-I		**Column-II**
(A)	Estimation of nitrogen	(p)	Chromato graphy
(B)	Estimation of halogen	(q)	Fractional distillation
(C)	Separate fractions of crude oil	(r)	Carius method
(D)	Separation of coloured substance found in plant	(s)	Duma's method

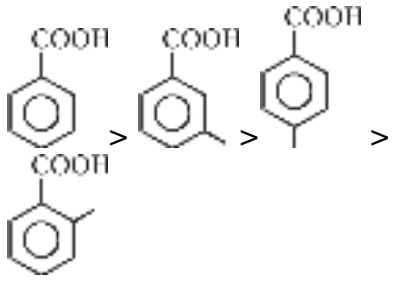
(1) A-s, B-r, C-q, D-p

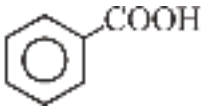
(2) A-r, B-s, C-q, D-p

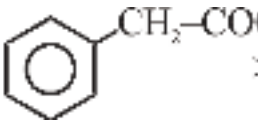
(3) A-s, B-r, C-p, D-q

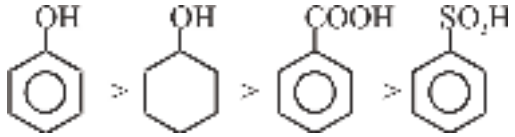
(4) A-q, B-p, C-r, D-s

25) Which is correct acidic order

(1)  >

(2) H–COOH >  > CH_3–COOH

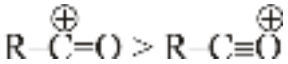
(3) CH_3–COOH >  > CH_3–CH_2–COOH

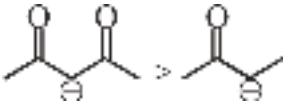
(4) 

26) Which of the following statement is incorrect ?

(1) Acidic strength - $CHCl_3$ > CHF_3

(2) Basic strength - Aniline > Pyrrole

(3)  (stability of resonating structure)

(4)  (stability order of carbanion)

27) Which of the following statement is correct for the given compounds ?

CH_3 – CH = CH_2 (a) CH_3 – C ≡ CH (b)

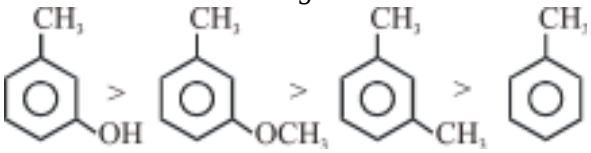
(1) (a) is more reactive than (b) for electrophilic addition reaction.

(2) Both (a) and (b) reacts with Tollen's reagent.

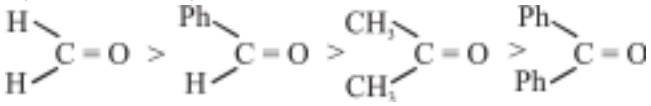
(3) (b) can be converted into (a) by reacting with Na/Liq. NH_3.

(4) All of the above

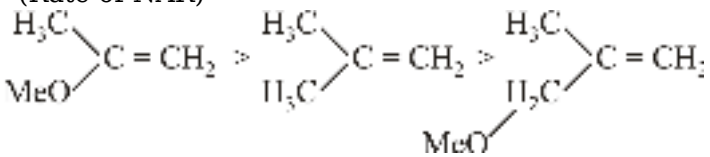
28) Which of the following order is correct :-

(a) 

(Rate of ESR)

(b) 

(Rate of NAR)

(c) 

(Rate of EAR)

(1) a, b, c

(2) only b, c

(3) only a, c

(4) only a, b

29) ***Assertion (A) :-*** Benzene reacts with excess of Cl_2 in presence of sunlight to form BHC.
Reason (B) :- Reaction undergoes electrophilic substitution.

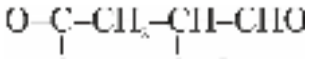
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

30)

The correct IUPAC name of the following compound  is :-

(1) 3,3–Diformyl propanoic acid

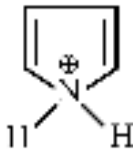
(2) 3–Formyl–4–oxobutanoic acid

(3) 3,3–Dioxo propanoic acid

(4) 3,3–Dicarbaldehyde propanoic acid

31) Which is incorrect statement -

(1) Hybridisation of nitrogen in pyrrole is sp^2

(2)  is Antiaromatic

(3) Halogens have –I effect and +M effect both but –I > +M.

(4) In Benzaldehyde (Ph–CHO), –M of –CHO affect only ortho and para position.

32) Which of the following reaction does not give ethane as product ?

(1) $CH_3\text{–}Cl \xrightarrow[\text{dry ether}]{Na}$

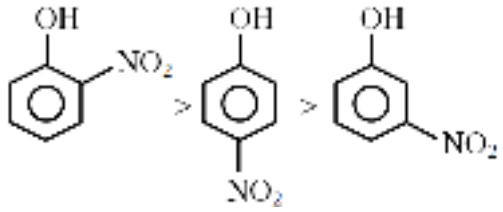
(2) $CH_3\text{–}COOH \xrightarrow[A]{\text{soda lime}}$

(3) $CH_3\text{–}COOK \xrightarrow{\text{electrolysis}}$

(4) $C_2H_5\text{–}Cl + H_2 \xrightarrow{Zn,H^{\oplus}}$

33)

Incorrect order of acidic strength is :-

(1) 

(2) $F_3C–COOH > F_2CH–COOH > F–CH_2–COOH$

(3) $H–COOH > Ph–COOH > CH_3COOH$

(4) $Ph–OH > C_2H_5OH > CH{\equiv}CH$

34) **Assertion (A):** If benzene is treated with 1-chloropropane in presence of anhydrous $AlCl_3$, Isopropyl benzene is major product not n-propyl benzene.
Reason (R) : Reaction occurs via rearrangement of carbocation.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

35) **Statement-I :** Hydrogenation of 2-Butyne in presence of $Pd/BaSO_4$ gives trans-2-Butene
Statement-II : Trans-2-butene is more stable than cis-2-butene

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

36) Total number of aromatic compounds/species among following are :-



(1) 5

(2) 2

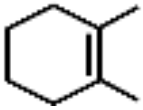
(3) 6

(4) 4

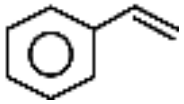
37)

Which compound will not form same major product with HBr and HBr/H_2O_2 ?

(1) 

(2) 

(3) 

(4) 

38) P and Q are :

$CH_3–C \equiv C–CH_3 + H_2 \xrightarrow[\text{quinoline}]{Pd/C} P$

$CH_3 – C \equiv C – CH_3 \xrightarrow[Na/Liq.NH_3]{} Q$

(1) Position isomers

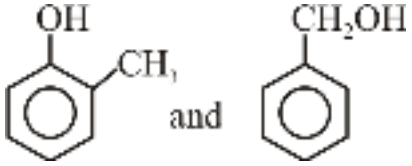
(2) Identical

(3) Geometrical isomers

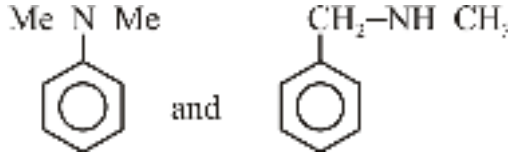
(4) None of the above

39)

Number of correct statement in following :

(a) 

Functional group isomers

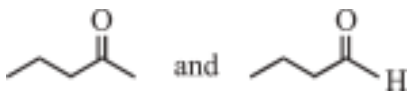
(b) 

Metamers

(c) $CH_3–C\equiv C–CH_3$ and 
Ring chain isomer

(d) 

Chain isomers

(e) 

Functional group isomers

(1) 2

(2) 5

(3) 4

(4) 3

40) Chloroform (B.P. 334K) and aniline (B.P. 457K) are easily separated by the technique of

(1) Crystallisation
(2) Distillation
(3) Sublimation
(4) Chromatography

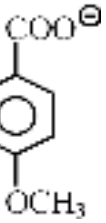
41) In detection of sulphur, sodium fusion is treated with lead acetate, then which colour is precipitated?

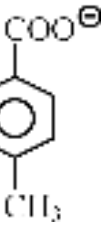
(1) White
(2) Yellow
(3) Black
(4) Violet

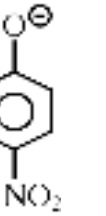
42) **Statement-I :-** Nitrogen, sulphur, halogen and phosphorous present in organic compound are detected by "Lassiagene test".
Statement-II :- In Lassigne test of nitrogen black precipitate indicates the presence of nitrogen.

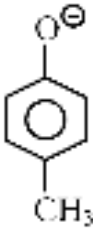
(1) Both statement I and statement II are true.
(2) Both statement I and statement II are false.
(3) Statement-I is true but Statement-II is false.
(4) Statement-I is false but Statement-II is true.

43) Which of the following is most stable anion.

(1) 

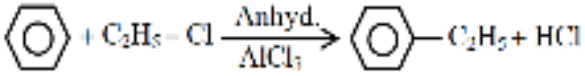
(2) 

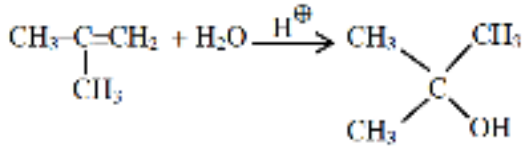
(3) 

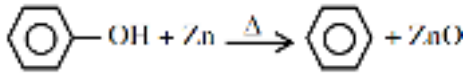
(4) 

44)

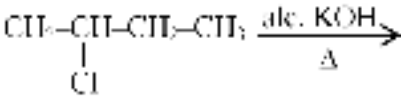
In which of the following reactions, formed product is incorrect ?

(1) 

(2) 

(3) 

(4) $CH_3-C\equiv CH + H_2O \xrightarrow[333\ K]{Hg^{+2}/H^{\oplus}} CH_3-CH_2-\underset{\underset{O}{\|}}{C}-H$

45)  Major product

Incorrect statement for above reaction.

(1) Product will show hyperconjugation effect.

(2) Product will show geometrical isomerism.

(3) Given reaction is dehydration.

(4) Given reaction is β-elimination.

BIOLOGY

1) Who provide evidence for production of glucose ?

(1) Preistley

(2) Ingenhousz

(3) Robert Hill

(4) von Sachs

2) There is a clear division of labour within the chloroplast. The membrane system is responsible for :-

(1) Trapping of chemical energy

(2) Formation of glucose

(3) Trapping of light energy

(4) Providing site for dark reaction

3) The graphic curve depicting the relative rates of photosynthesis at different wave lengths of light is called :

(1) Absorption spectrum

(2) Action spectrum

(3) Atomic spectrum

(4) Emission spectrum

4) The single chlorophyll-a molecule having absorption peak at 680 nm, forms the :-

(1) Reaction centre of PS I

(2) Reaction centre of PS II

(3) Antennae of PS I

(4) Antennae of PS II

5) Which of the following is/are mobile carriers in the thylakoid membrane ?

(1) Plastoquinone

(2) Cytochrome b_6f complex

(3) Plastocyanin

(4) Both (1) and (3)

6) Micronutrient manganese is essential for which of the following important event of photosynthesis ?

(1) Cyclic photophosphorylation

(2) Non-cyclic photophosphorylation

(3) Photolysis of water

(4) Biosynthetic phase

7) **Statement-I :** Within the chloroplast, protons in the stroma decrease in number, while in lumen there is accumulation of protons.
Statement-II : This creates proton gradient across the thylakoid membrane as well as measurable increase in pH in the lumen.

(1) Both statements I and II are correct

(2) Statement-I is correct and statement-II is incorrect.

(3) Statement-I is incorrect and statement-II is correct

(4) Both statements I and II are incorrect.

8) Which statement is incorrect for biosynthetic phase of C_3 plants ?

(1) 1st stable product is 3-phosphoglyceric acid

(2) Primary acceptor of CO_2 is oxaloacetic acid.

(3) Primary acceptor of CO_2 is RuBP

(4) ATP and NADPH are used to drive the process.

9) Given below are two statements : One is labelled as Assertion (A) and other is labelled Reason (R).
Assertion (A) :- In C_4 plants, the bundle sheath cells may form several layers around the vascular bundles, they are characterised by having a large number of chloroplast, thick walls impervious to gaseous exchange & no intercellular spaces.
Reason (R) :- The mesophyll cell lack RubisCO enzyme but have PEP carboxylase.
In the light of the above given statements, choose the correct answer from the options given below :-

(1) A and R both are true and R is the correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) A and R both are true but R is not correct explanation of A.

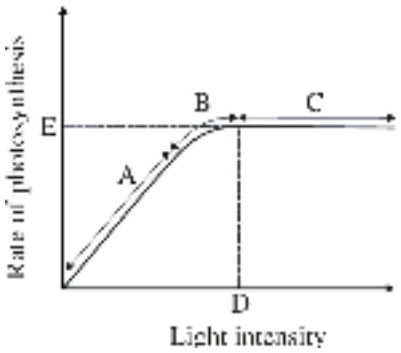
10) Read the following statements given below :
(a) RuBisCO is the most abundant enzyme in the world.
(b) RuBisCO has a much greater affinity for O_2 when O_2 : CO_2 is nearly equal.
(c) In photorespiratory pathway, there is synthesis of sugars but not of ATP.
(d) In C_4 plants photorespiration does not occur.
How many of above statements are correct :

(1) a & b only

(2) a, b, c, d

(3) a & d only

(4) a, c & d only



11)
Select the correct option about given diagram :-

(1) A - Light intensity is limiting
B - Other factors became limiting

(2) A - Other factors became limiting
B - Optimum rate of photosynthesis

(3) C - Other factors became limiting
D - Light saturation point

(4) E - Photosynthesis saturation point
B - Light saturation point

12) Match the column - I with column - II & select the correct option.

Column - I		Column - II	
A.	Site of bacterial photosynthesis	I.	Thylakoid
B.	Site of pigments in green plants	II.	Cytoplasm
C.	Site of calvin cycle in C_4 plants	III.	Mesophyll cell
D.	Site of light reaction in C_4 plants	IV.	Bundle sheath cell

(1) A – II, B – I, C – III, D – IV

(2) A – II, B – I, C – IV, D – III

(3) A – II, B – IV, C – I, D – III

(4) A – III, B – I, C – II, D – IV

13) **Statement-I :** In C_4 plant oxygenase activity of RuBisCo is maximised due to increased O_2 concentration in mesophyll cell.
Statement-II : In C_3 plant no O_2 can bind to RuBisCo because the enzyme RuBisCo is present in its bundle sheath cell.
Select the correct option :-

(1) Both the statements are correct.

(2) Both the statements are incorrect.

(3) Statement-I correct and statement-II incorrect

(4) Statement-I incorrect and statement-II correct

14) **Statement-I :** FNR takes/receives e^- and H^+ from the PS-I and reduces the $NADP^+$.
Statement-II : Photolysis of H_2O, proton pumping and activity of FNR are responsible for decrease in pH of lumen.
Select the correct option :-

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement-I correct and statement-II incorrect.

(4) Statement-I incorrect and statement-II correct.

15) Match the following :-

	List-I		List-II
(a)	T.W Englemann	(i)	Plants restore to the air whatever breathing animals and burning candles remove.
(b)	Jan Ingenhousz	(ii)	Only the green part of the plants can release oxygen
(c)	Julius Van Sachs	(iii)	Green parts of the plants form glucose
(d)	Joseph Priestly	(iv)	First action spectrum in photosynthesis

(1) a-iv, b-iii, c-i, d-ii

(2) a-iii, b-iv, c-ii, d-i
(3) a-iv, b-iii, c-ii, d-i
(4) a-iv, b-ii, c-iii, d-i

16) Which of the following structure or process is not related to gaseous exchange in plants.

(1) Stomata
(2) Lenticels
(3) Diffusion
(4) Osmosis

17) After glycolysis, what is the fate of pyruvic acid:

(1) Krebs cycle entry
(2) Alcoholic fermentaion
(3) Lactic acid fermentation
(4) All of the above

18) Which of the following scientist elucidated TCA cycle ?

(1) J. parnas
(2) Embden
(3) Darwin
(4) Hans Krebs

19) In mitochondrial E.T.S complex-I is also called :-

(1) FADH dehydrogenase
(2) Cytochrome 'C' oxidase
(3) NADH dehydrogenase
(4) ATP synthase

20) During fermentation, what is the net gain of ATP by oxidative phosphorylation when using glucose as a respiratory substrate ?

(1) 2
(2) 6
(3) 8
(4) 0

21) How do you think alcoholic beverages of alcohol content greater than 13% concentrations are obtained ?

(1) By fermentation
(2) By distillation
(3) By malting

(4) By coagulation

22) Which of the following statement are correct ?
Choose the correct answer from the option given below :
(A) Enzyme for glycolysis is present in cytoplasm.
(B) Glycolysis occur only in eukaryotes.
(C) Pyruvic acid is the key product of glycolysis.
(D) In plants, glucose is derived from sucrose which is the end product of photosynthesis.
(E) Sucrose is converted into glucose and fructose by enzyme invertase.

(1) A, C, D only
(2) A, B, D only
(3) A, C, E only
(4) A,C, D & E only

23) Pyruvic acid, which is formed by glycolysis in cytosol, after it enters mitochondrial matrix undergoes ___A___ by a complex set of reactions catalysed by ___B___.

(1) A = Oxidative phosphorylation, B = Lactate dehydrogenase
(2) A = Dephosphorylation, B = Pyruvate kinase
(3) A = Oxidative decarboxylation, B = Pyruvate kinase
(4) A = Oxidative decarboxylation, B = Pyruvate dehydrogenase

24) **Statement-I :-** During the conversion of succinyl CoA to succinic acid a molecule of GTP is synthesised.
Statement-II :- In a coupled reaction GTP is converted to GDP with a simultaneous synthesis of ATP from ADP.

(1) Statement I is correct & statement II is incorrect
(2) Statement I is incorrect & statement II is correct
(3) Both Statement I & statement II are correct
(4) Both Statement I & statement II are incorrect

25) Match column-I & column-II.

Column-I		**Column-II**	
(i)	Complex-I	(a)	Cyt b, cyt c, FeS
(ii)	Complex-II	(b)	FMN, FeS
(iii)	Complex-III	(c)	FAD, FeS
(iv)	Complex-IV	(d)	Cyt a, cyt a_3, Cu_A, Cu_B

(1) i-b, ii-a, iii-c,iv-d
(2) i-b, ii-d, iii-c,iv-a
(3) i-b, ii-c, iii-a,iv-d
(4) i-b, ii-d, iii-a,iv-c

26) **Assertion :** Animal muscle cells produce lactic acid during exercise.
Reason : When oxygen is inadequate for cellular respiration pyruvic acid is reduced to lactic acid.

(1) Both **Assertion** and **Reason** are correct and **Reason** is correct explanation of **Assertion**.

(2) **Assertion** is correct but **Reason** is incorrect

(3) **Assertion** is incorrect but **Reason** is correct

(4) Both **Assertion** and **Reason** are correct and **Reason** is not correct explanation of **Assertion**.

27) Which of the following statement is incorrect :-

(1) Respiratory pathway is involved in both anabolism and catabolism it would be better to consider the respiratory pathway as an amphibolic pathway

(2) Fatty acid would enter the pathway after being converted to PGAL.

(3) Glycerol would enter the pathway after being converted to PGAL.

(4) Fatty acid would be broken down to Acetyl CoA before entering the respiratory pathway.

28) There are two columns 1 and column 2. Match the columns with value of RQ and substrate is used.

	Column-I		**Column-II**
(A)	Tripalmitin	(i)	1.0 RQ
(B)	Pure proteins and fats	(ii)	0.7 RQ
(C)	Glucose	(iii)	Not used as respiratory substrate
(D)	OAA (Oxalo acidic acid)	(iv)	4.0 RQ

	A	B	C	D
1.	ii	iii	i	iv
2.	i	iii	iv	ii
3.	iii	i	ii	iv
4.	ii	iv	iii	i

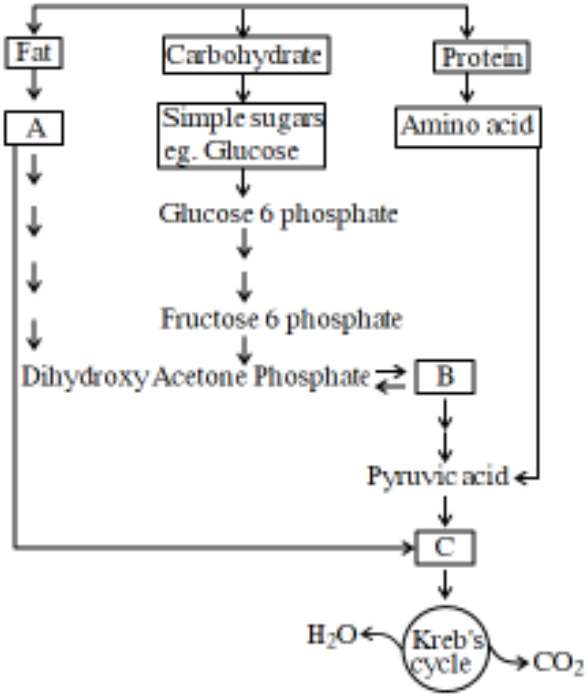
(1) 1

(2) 2

(3) 3

(4) 4

29) Given below is the interrelationship among metabolic pathway showing respiration mediated breakdown of different organic molecules to CO_2 and H_2O.



Identify A,B and C

(1) A = Acetyl CoA
B = Glyceraldehyde 3 phosphate
C = Fatty acid and glycerol

(2) A = Glyceraldehyde 3 phosphate
B = Acetyl CoA
C = Fatty acid and glycerol

(3) A = Fatty acid and glycerol
B = Glyceraldehyde 3 phosphate
C = Acetyl CoA

(4) A = Dihydroxy acetone phosphate
B = Fatty acid and glycerol
C = Glyceraldehyde 3 phosphate

30) **Statement-I :** Ubiquinone receives reducing equivalents via $FADH_2$ that is generated during oxidation of succinate in TCA cycle.
Statement-II : Ubiquinone is oxidised with the transfer of electron to cytochrome bc_1 complex.

(1) Both Statement I and Statement II are incorrect.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are correct.

31) Single maize root apical meristem can given rise to :-

(1) 17,500 cells per minute
(2) 3,50000 cells per hour
(3) 1750 cells per second
(4) 17500 cells per hour

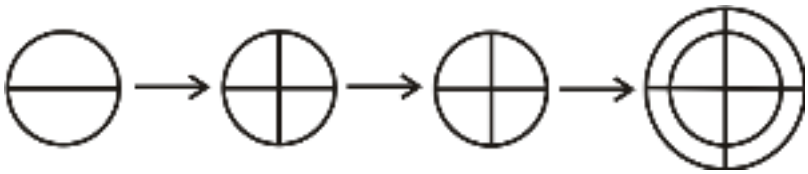
32) Which of the following is not a characteristic of elongation phase ?

(1) New cell wall deposition

(2) Cell wall thickening

(3) Cell enlargement

(4) Increased vacuolation

33) 

Above figure represents :-

(1) Arithmetic growth phase

(2) Geometric growth phase

(3) Arithmetic and geometric phase

(4) Arithmetic growth phase, cells that lode the capacity of division.

34) Which of the following is not an extrinsic factor ?

(1) Light, Temperature, Water

(2) Light, Oxygen, Nutrition

(3) Nutrition, PGR, Gene

(4) Oxygen, Water, Temperature

35) In tissue culture experiments, parenchymatous cells are put in a culture medium to form callus, this phenomenon may be called as :-

(1) Senescence

(2) Development

(3) Dedifferentiation

(4) Differentiation

36) Which of the following is an example of plasticity :-
(i) Heterophylly in buttercup
(ii) Homophylly in cotton
(iii) Heterophylly in larkspur
(iv) Heterophylly in coriander

(1) (i), (ii), (iii), (iv)

(2) (iii), (iv) only

(3) (i), (iii), (iv)

(4) (i), (iii) only

37) Read the following terminology of column I & II and identify the correct match :

	Column-I		**Column-II**
(A)	Indole derivatives	(i)	ABA

(B)	Adenine derivatives	(ii)	Ethylene
(C)	Carotenoids derivatives	(iii)	IAA
(D)	Terpenes derivatives	(iv)	Kinetin
		(v)	Gibberellic acid

(1) A-(iii), B-(iv), C-(i), D-(v)

(2) A-(iii), B-(iv), C-(v), D-(iv)

(3) A-(i), B-(ii), C-(iii), D-(iv)

(4) A-(i), B-(ii), C-(iv), D-(iii)

38) **Statement-I :-** Miller et. al identified and crystallised the cytokinesis promoting active substance that they termed kinetin.
Statement-II :- Inhibitor B, abcission II and dormin all the three were proved to be chemically identical it was named abscisic acid.

(1) Both Statement are true

(2) Both Statement are false

(3) Statement I true and Statement II false

(4) Statement I false and Statement II true

39) Auxin used by gardeners to prepare weed free lawns. Select the incorrect statement about it.

(1) It is a selective weed killer.

(2) 2, 4-D is widely used to kill dicotyledonous weeds.

(3) It is used to kill monocotyledonous weeds.

(4) It does not affects mature monocotyledons plants.

40) Gibberellins do not cause :-

(1) Promotion of Parthenocarpy

(2) Stimulation of seed germination

(3) GA_3 is used as to speed up malting process

(4) Shortening of genetically tall plant

41) __A__ plant hormone promote nutrient mobilisation which helps in the delay of leaf senescence. Identify the A :

(1) Auxin

(2) Gibberellins

(3) Cytokinin

(4) Ethylene

42) Which plant hormone is responsible for femaleness (Ferninising effect) in cucumbers ?

(1) Abscisic acid

(2) Cytokinine

(3) Ethylene

(4) Auxin

43) ABA plays a major role in :-

(1) seed development

(2) seed maturation

(3) seed dormancy

(4) All of the above



44) The diagram represents :-

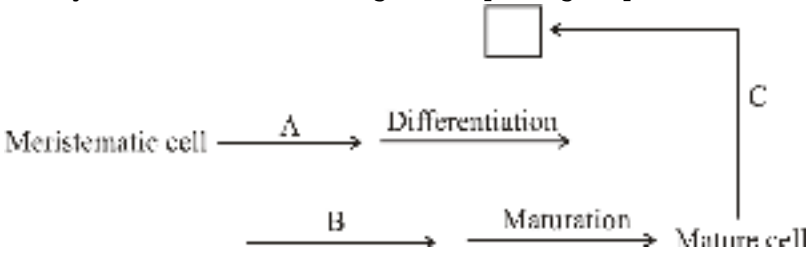
(1) (a) Water habitat
(b) Heterophylly
(c) Larkspur

(2) (a) Water habitat
(b) Heterophylly
(c) Cotton

(3) (a) Terrestrial habitat
(b) Heterophylly
(c) Buttercup

(4) (a) Water habitat
(b) Heterophylly
(c) Buttercup

45) Identify A, B and C in the diagram depicting sequence of the development process in a plant cell:-



	A	B	C
(1)	Cell division	Expansion	Senescence
(2)	Cell growth	Plasmatic growth	Expansion
(3)	Plasmatic growth	Expansion	Senescence
(4)	Senescence	Expansion	Plasmatic growth

(1) 1

(2) 2

(3) 3

(4) 4

46) Match the columns and select the correct option :

Column-I		Column-II	
A.	Corpus callosum	i.	Body temperature
B.	Thalamus	ii.	Connect cerebral hemisphere
C.	Hypothalamus	iii.	Covering of brain
D.	Arachnoid	iv.	Major coordinating centre for sensory and motor signaling

Options :-

(1) A-ii, B-i, C-iv, D-iii

(2) A-iii, B-ii, C-i, D-iv

(3) A-ii, B-iv, C-i, D-iii

(4) A-iv, B-ii, C-i, D-iii

47) Transmission of an impulse across electrical synapses is :-

(1) Similar to chemical synapse

(2) Very similar to impulses conduction along single axon

(3) Very slow

(4) Depend upon neurotransmitters

48) When the neuron is conducting an impulse, i.e. excited, the axon membrane :

(1) Is freely permeable to sodium and potassium ions

(2) Comparatively more permeable for potassium and nearly impermeable to sodium ions

(3) Rise in permeability to potassium ions

(4) Becomes negatively charge outside and positively charge inside

49) Which type of neuron found in the retina of eye and cerebral cortex, respectively?

(1) Multipolar neuron and Bipolar neuron

(2) Unipolar neuron and Multipolar neuron

(3) Bipolar neuron and Multipolar neuron

(4) Pseudounipolar and Multipolar neuron

50)

Which of the following is not correct regarding to neurons ?

(1) Neurons can detect, receive and transmit different type of stimules.

(2) In Hydra nervous system is composed of network of neurons

(3) Vertebrates have less developed neural system.

(4) Neurons are highly specialised cells.

51) **Assertion (A):** Myelinated nerve fibre are enveloped with schwann cell, which form a myelin sheath around the axon.
Reason (R): Unmylinated nerve fibre is not enclosed by a schwann cells.

(1) Both A & R are correct but R is not correct explaination of A

(2) Both A & R are correct and R is correct explaination of A

(3) A is correct but R is not correct

(4) Both A & R are incorrect

52) Given below are two statements:
Statement I : The neural organisation in Hydra is composed of a network of neurons. The neural system is better organised in insects, where a brain is present along with a spinal cord and neural tissues. The vertebrates have a more developed neural system.
Statement II : The neural system of all animals is composed of highly specialised cells called neurons which can detect, receive and transmit different kinds of stimuli.
In the light of above Statement choose the correct answer from the options given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

53) Read following terminology of column-I and column-II and identify their correct match :-

Column-I		**Column-II**	
(A)	Cerebrum	(i)	Balance and posture
(B)	Hypothalamus	(ii)	Memory
(C)	Medulla	(iii)	Thermoregulation
(D)	Cerebellum	(iv)	Controls respiration
		(v)	Controls cardiovascular reflex
		(vi)	Controls endocrine system
		(vii)	Coordination of voluntary muscles

(1) (A)-(ii), (B)-(iv),(v), (C)-(iii),(vi), (D)-(i),(vii)

(2) (A)-(ii), (B)-(iii),(vi), (C)-(iv),(v), (D)-(i),(vii)

(3) (A)-(i),(vii), (B)-(iii),(vi), (C)-(iv),(v), (D)-(ii)

(4) (A)-(i), (B)-(ii),(iii), (C)-(iv),(v), (D)-(vi),(vii)

54) Read the following statements and choose the correct option -

(1) Chemical synapse are faster than electrical.

(2) Electrical synapse are rare in our body.

(3) Rise in the stimulus induced permeability to K^+ is extremely long lived.

(4) Resting potential is also known as nerve impulse.

55) **Statement-I :** Nerve impulse passes along the axon as a wave of depolarization.
Statement-II : In depolarization, Na^+ diffuse outside the membrane at the site of excitation.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Only Statement-II is correct

56) The axonal membrane is _____ to negatively charged proteins present in the axoplasm.

(1) Selectively permeable

(2) permeable

(3) Semipermeable

(4) Impermeable

57) Find the incorrect statements :-
(a) The resting potential is maintained by sodium potassium pump.
(b) Across the cell surface membrane of nerve cells, positive charge inside the cell with respect to outside is said to be polarised.
(c) Rapid influx of K^+ ions leads action potential.
(d) Na^+- K^+ pump cause removal of $3Na^+$ and uptake of $2K^+$

(1) a and b

(2) b and c

(3) c and d

(4) a and d

58) Glucocorticoids stimulates :-

(1) Gluconeogenesis

(2) Lipolysis

(3) Proteolysis

(4) All of these

59) Select the hormones which interact with intracellular receptors :-

(1) Epinephrine and cortisol

(2) Estradiol and Insulin

(3) Testosterone and glucagon

(4) Thyroid hormone and Estradiol

60) Underproduction of which of the following hormone(s) results in excretion of large amounts of Na^+ in human urine?

(1) Mineralocorticoids
(2) Glucagon
(3) Thyroid
(4) FSH

61) Name of a peptide hormone secreted by pumping station of body, which decrease blood pressure:-

(1) Atrial Natriuretic factor
(2) Aldosterone
(3) Vasopressin
(4) GIP

62) Identify the correct statement :

(1) Hyperthyroidism during pregnancy results in growing baby to (cretinism) stunted growth
(2) PTH is hypocalcemic hormone
(3) Hyperglycemia results due to release of glucagon
(4) Hyper-production of adrenal cortex hormone leading to a disease called addison's disease

63) **Statement-I :-** Over secretion of GH stimulates abnormal growth of the body leading to dwarfism.
Statment-II :- LH and FSH stimulates gonadal activity and hence are called gonadotrophins.

(1) Statement I and II correct
(2) Statement I and II incorrect
(3) Statement I correct and II incorrect
(4) Statement I incorrect and II correct

64) Match the source gland with respective hormone as well as the function :-

	Source gland	Hormone	Function
(1)	Anterior pituitary	Oxytocin	Contraction of uterus muscles during child birth
(2)	Posterior pituitary	Vasopressin	Stimulates re-absorption of water in the distal tubules in the nephron
(3)	Corpus luteum	Estrogen	Increase secretory nature of Endomatrium
(4)	Thyroid	Thyroxine	Regulates blood calcium level

(1) 1

(2) 2

(3) 3

(4) 4

65) Match List I with List II and select the correct option :-

	List I		List II
A	Adrenaline	1	Goitre
B	Hyperparathyroidism	2	Accelerates heart beat
C	Oxytocin	3	Salt-water balance
D	Hypothyroidism	4	Childbirth
E	Aldosterone	5	Demineralisation

(1) A-2, B-5, C-4, D-1, E-3

(2) A-3, B-4, C-5, D-3, E-2

(3) A-5, B-3, C-2, D-4, E-1

(4) A-2, B-3, C-4, D-5, E-1

66)

Which of the following are not the effects of insulin hormone ?
(a) Stimulates glycogenesis
(b) Increases glucose level in blood
(c) Increases the cellular glucose uptake
(d) deficiency leads to Diabetes insipidus
(e) Stimulates glycogenolysis
Choose the **most appropriate** answer from the options given below :

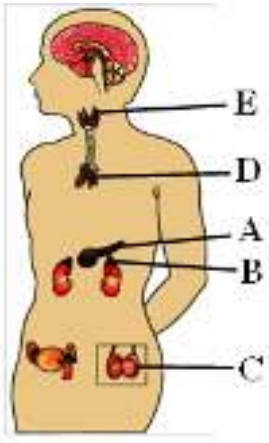
(1) (b), (d) and (e) only

(2) (a) and (e) only

(3) (b) and (c) only

(4) (a) and (c) only

67)



In the above shown diagram, how many are organised endocrine glands ?

(1) Four

(2) Three

(3) Two

(4) Five

68) Some events about gonadotrophin hormones. secretions are given below, read the carefully and arranged them in proper sequence.
A. Secretion of FSH and LH from anterior pituitary.
B. Transport of GnRH through portal circulation from hypothalamus to anterior pituitary.
C. Secretion of GnRH from hypothalamus.
D. Stimulation of gonadal activity.
E. Stimulation of anterior pituitary.

(1) C → B → E → D → A

(2) C → B → E → A → D

(3) C → E → A → D → B

(4) C → A → D → E → B

69) **Assertion:** The hormones produced by hypothalamus are of two types, the releasing hormones and the inhibiting hormones.
Reason: A hypothalmic hormone called Gonadotrophin releasing hormone(GnRH) stimulates the pituitary gland to synthesis and release of gonadotrophins. On the other hand, somatostatin from the hypothalamus inhibit the growth hormone from the pituitary.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

70) Which of the following statement is incorrect?

(1) Testis and ovary function as primary sex organ as well as endocrine gland

(2) Large amount of androgenic steroids is secreted by adrenal cortex

(3) The leydig's cell secrete testosterone under the influence of ICSH

(4) Glucagon reduces the cellular glucose uptake and utilization

71) Match the followings and choose the correct options -

Hormone		**Function**	
A.	Vasoprssin	i	Milk ejection
B	Oxytocin	ii	Reduces loss of H_2O in urine
C	Prolactin	iii	Alveolar growth in pregnancy
D	Progesteron	iv	Milk formation

(1) A – ii, B – iii, C – iv, D – i
(2) A – ii, B – iii, C – i, D – iv
(3) A – ii, B – iv, C – i, D – iii
(4) A – ii, B – i, C – iv, D – iii

72)

Which of the following stimulates glycogenolysis-

(1) Catecholamines and cortisol
(2) Cortisol ans insulin
(3) ADH and cortisol
(4) Both 1 & 2

73)

Which of the following glands regulates blood Ca^{++} level-

(1) Parathyroid
(2) Thyroid
(3) Adrenal
(4) Both (1) & (2)

74)

Which of the following doesnot transport axonally to pars nervosa-

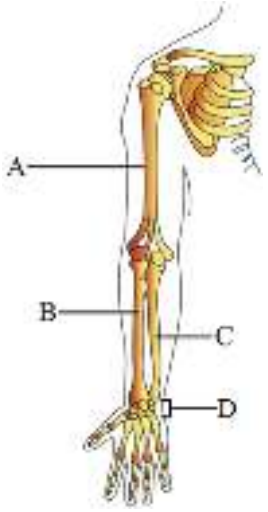
(1) Prolactin
(2) ADH
(3) Oxytocin
(4) Vasopressin

75) Thymus gland is located on which side of heart and aorta ?

(1) Ventral
(2) Dorsal
(3) Dorso-lateral
(4) Ventro-lateral

76) Somatostatin inhibits the secretion of ?

(1) Gonadotropins
(2) TSH
(3) Prolactin
(4) Growth hormone



77) Identify the given below diagram and match correct option :-

(1) A-Radius, B-Ulna, C-Humerus, D-Carpals

(2) A-Humerus, B-Radius, C-Ulna, D-Carpals

(3) A-Humerus, B-Ulna, C-Radius, D-Carpals

(4) A-Carpals, B-Ulna, C-Radius, D-Humerus

78) Following are the events of muscular contraction, arrange them in correct order?
(A) Action potential generates in the sarcolemma
(B) Hydrolysis of ATP
(C) Myosin head pulls the attached actin filaments
(D) Release of calcium ions into the sarcoplasm
(E) Shortenins of sarcomere

(1) A → B → C → D → E

(2) A → C → D → B → E

(3) A → D → B → C → E

(4) A → C → B → D → E

79)

Fibrous joint	Cartilagenous joint	Synovial joint
↓	↓	↓
(A)	(B)	(C)

Choose the correct option :-

(1) A → Saddle joint, B → Pubic-symphysis, C → Knee joint

(2) A → Hinge joint, B → Knee joint, C → Pubic-symphysis

(3) A → Sutures, B → Inter vertebral disc, C → Knee joint

(4) A → Sutures, B → Knee joint, C → Hinge joint

80)

How many of the following are examples of synovial joint ?

Knee joint, Pubic symphysis, Suture, Radioulnar joint, Hip joint, Ankle joint, Wrist joint, Intervertebral disc

(1) Four

(2) Five

(3) Six

(4) Three

81) Which of the following character represent skeletal muscle correctly ?
(A) Fascicles are held by fascia.
(B) Fascia is an elastic connective tissue layer.
(C) Sarcoplasmic reticulum is structural and functional unit of muscle contraction.
(D) M-line connects myosin filaments together.

(1) A and B only

(2) A and C only

(3) B and C only

(4) A and D only

82) Which is not an example of long bones :-

(1) Humerus

(2) Radius

(3) Ulna

(4) Carpals

83) Cross arm of myosin is made up of :-

(1) Head + tail

(2) Tail + short arm

(3) Head + short arm

(4) Only short arm

84) Intercalated discs are characterstics of muscles that are found in :-

(1) Brain

(2) Spinal cord

(3) Stomach

(4) Heart

85) Joint present between atlas and axis :-

(1) Saddle joint

(2) Pivot joint

(3) Gliding joint

(4) Hinge joint

86) Age related disorder characterised by decreased bone mass and increase chances of fractures is-

(1) Tetany

(2) Gout

(3) Arthritis

(4) Osteoporosis

87) **Assertion :** Muscle fibre is the anatomical unit of muscle.
Reason : Each muscle fibre has many parallelly arranged myofibrils.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

88) Given below two statements :-
Statement-I : Thoracic vertebrae, Ribs and Sternum together form the rib cage.
Statement-II : Last 2 pairs (11th and 12th) of ribs are not connected ventrally and are therefore called Floating ribs.
Choose the correct answer from the options given below :-

(1) Both Statement I and Statement II are false

(2) Statement I is true but Statement II is False

(3) Statement I is false but statement II is Ture

(4) Both Statement I and Statement II are True

89) Between humerus and pectoral girdle which type of joint is present :-

(1) Pivot joint

(2) Hinge joint

(3) Ball & socket joints

(4) Gliding joints

90) Cranium of man is made up of

(1) 8 bones

(2) 12 bones

(3) 16 bones

(4) 18 bones

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	3	4	2	1	2	3	3	2	1	1	2	3	3	2	4	2	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	2	3	2	3	3	2	4	1	3	4	1	1	2	2	2	2	4	4
Q.	41	42	43	44	45															
A.	3	2	3	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	2	2	3	3	4	4	2	3	4	2	2	3	3	2	4	3	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	2	1	2	3	1	1	2	2	2	2	1	2	4	4	4	3	1	2
Q.	86	87	88	89	90															
A.	3	3	2	4	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	2	2	4	3	2	2	4	3	3	2	2	1	4	4	4	4	3	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	4	4	3	3	1	2	1	3	2	4	2	2	3	3	3	1	1	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	4	3	3	3	2	4	3	3	3	3	2	2	3	4	2	4	4	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	3	4	2	1	1	4	2	1	2	4	1	4	1	1	4	2	3	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	3	4	2	4	2	4	3	1										

SOLUTIONS

PHYSICS

1) Restoring force is zero at mean position

$F = -Kx + F^0 \Rightarrow 0 = -Kx + F^0 \Rightarrow x = \frac{F_0}{K}$

i.e. the particle will oscillate about $x = \frac{F_0}{K}$

Also, $a = -\frac{K}{m}x + \frac{F_o}{m}$

$\therefore \omega = \sqrt{\frac{K}{m}}$

2) $y_1 = \frac{1}{2}\sin \omega t + \frac{\sqrt{3}}{2}\cos \omega t$

$A_1 = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1$

$\tan \alpha_1 = \sqrt{3} \Rightarrow \alpha_1 = \pi/3$

$y_1 = \sin(\omega t + \pi/3)$ ...(i)

$y_2 = \sin \omega t + \sin(\omega t + \pi/2)$

$\therefore A_2 = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\tan \alpha_2 = 1 \Rightarrow \alpha_2 = \pi/4$

$y_2 = \sqrt{2}\sin(\omega t + \pi/4)$ ...(ii)

phase diff. (from (i) & (ii))

$= \pi/3 - \pi/4 = \pi/12.$

3) $\theta = \theta_o \sin(\omega t + \phi)$

$T = 2\pi\sqrt{\frac{L}{g}} \approx 2\text{sec}$

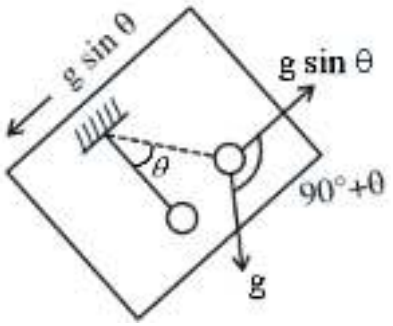
4) $\frac{1}{2}KA^2 = \frac{1}{2}K(0.2)^2 = 1$

$K = \frac{2}{(0.2)^2} = \frac{2 \times 100}{4} = 50$

$\omega = \sqrt{K/m}$

$= \sqrt{\frac{50}{4}} = \frac{5}{2}\sqrt{2} = \frac{5}{\sqrt{2}}$

$\boxed{T = \frac{2\pi}{\omega} = \frac{2\sqrt{2}\pi}{5}\text{s}}$



5)

$$T = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$$

$$g_{eff} = \sqrt{g^2 + g^2\sin^2\theta + 2(g)(g\sin\theta)\cos(90^\circ + \theta)}$$

$$= \sqrt{g^2 + g^2\sin^2\theta - 2g^2\sin^2\theta}$$

$$= \sqrt{g^2(1-\sin^2\theta)}$$

$$g_{eff} = \sqrt{g^2\cos^2\theta} = g\cos\theta$$

$$T = 2\pi\sqrt{\frac{\ell}{g\cos\theta}}$$

6) For a simple pendulum $T = 2\pi\sqrt{\frac{L}{g_{eff}}}$

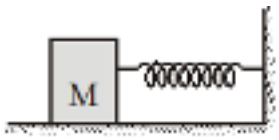
situation 1 : when pendulum is in air → $g_{eff} = g$

situation 2 : when pendulum is in liquid

→ $g_{eff} = g\left(1 - \frac{\rho_{liquid}}{\rho_{body}}\right) = g\left(1 - \frac{1}{16}\right) = \frac{15g}{16}$

$$\left(1 - \frac{\rho_{liquid}}{\rho_{body}}\right) = g\left(1 - \frac{1}{16}\right) = \frac{15g}{16}$$

So, $\frac{T'}{T} = \frac{2\pi\sqrt{\frac{L}{15g/16}}}{2\pi\sqrt{\frac{L}{g}}} \Rightarrow T' = \frac{4T}{\sqrt{15}}$

7) 

Momentum of system remains conserved.

$p_i = p_f$

$MA\omega = (m + M)\, A'\omega'$

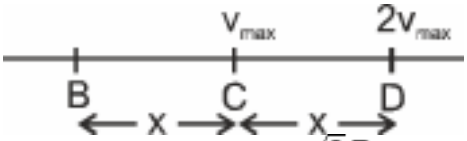
$MA\sqrt{\frac{k}{M}} = (m + M)\, A'\sqrt{\frac{k}{m+M}}$

$A' = A\sqrt{\frac{M}{M+m}}$

8) When the particle crosses D, its speed is half the maximum speed.

$\Rightarrow v = \frac{v_{max}}{R}\sqrt{R^2 - x^2}$

or $\frac{v_{max}}{2} = \frac{v_{max}}{R}\sqrt{R^2 - x^2}$ or $x = \frac{\sqrt{3}}{2} R$



$\Rightarrow$ Distance BD = $2x = \sqrt{3}R$

9) Conservation of momentum when collision take place mv = (M + m)(Aω)
{Aω = maximum velocity in SHM}

$$A = \frac{mv}{(m+M)\omega} = \frac{mv}{(m+M)\sqrt{\frac{k}{m+M}}}$$

$$A = \frac{mv}{\sqrt{(m+M)k}}$$

10)

Let total mass initially: $m_1 + m_2 + m_3 = 700 + 500 + 400 = 1600\ g$
After removing $700\ g$ mass:

$$T_1 = 3\ s = 2\pi\sqrt{\frac{900}{k}} \quad .....(1)$$

After removing 500 g mass, remaining mass is 400g

$$T_2 = 2\pi\sqrt{\frac{400}{k}} \quad .....(2)$$

Now, equation (2) divided by equation (1)

$$\frac{T_2}{3} = \frac{20}{30}$$

Then,

$$T_2 = 2\ s$$

11) Energy is lost due to work done by non conservative force.

12) λ = 2[0.49 – 0.16]
λ = 2[0.33] m

$$n = \frac{v}{\lambda} = \frac{330}{2 \times 0.33} = 500\,Hz$$

13) $f = \frac{v}{2\ell} = 400\ Hz$

$$f' = \frac{v}{4\left(\frac{\ell}{2}\right)} = 400\ Hz$$

14)

Given $\frac{\text{Mass}}{\text{Vol.}} = 1$

$\frac{M}{A\ell} = 1$

$\frac{M}{\ell} = \mu = A = 0.01 \quad \left[\because V = \frac{w}{k} = \frac{20}{10} = 2\,m/s\right]$

$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{mg}{\mu}}$

$2 = \sqrt{\frac{m(10)}{0.01}} \Rightarrow \boxed{m = 0.004\,kg}$

15)

Phase difference = $\frac{2\pi}{\lambda}$ × path difference

⇒ 1.6π = $\frac{2\pi}{\lambda}$ × 40 ⇒ λ = 50 cm = 0.5 m

⇒ v = nλ ⇒ 330 = 0.5 × n ⇒ n = 660 Hz

16) $v_{rms} = \sqrt{\frac{3RT}{M}}$

$v_{sound} = \sqrt{\frac{\gamma RT}{M}}$

$\frac{v_{rms}}{v_{sound}} = \sqrt{\frac{3}{\gamma}}$

equation of the graph $\left(\frac{1}{V}\right) \propto T^2 \quad \left[x \propto y^2\right]$

⇒ T^2V = constant (process equation)

⇒ $PV^{3/2}$ = constant ⇒ $\gamma = \frac{3}{2}$

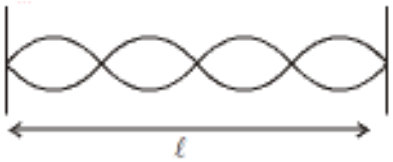
17) Given, temperature of the source T_1 = 500 K
Temperature of the sink, T_2 = 300 K
Work done per cycle, W = 1 KJ = 1000 J
Heat transferred to the engine per cycle, Q_1 =?
Efficiency of a Carnot engine,

$$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{500} = \frac{2}{5}$$

and $\eta = \frac{W}{Q_1}$

or $Q_1 = \frac{W}{\eta} = \frac{1000}{(2/5)} = 2500\,J$

18) 4th harmonic



$4\frac{\lambda}{2} = \ell$

$2\lambda = \ell$

From equation, $\frac{2\pi}{\lambda} = 0.157$

$\lambda = 40$ m

$\ell = 2\lambda = 80$ m

19) $n_b = n_1 - n_2$

$T_b = \frac{1}{n_b} = \frac{1}{n_1 - n_2}$

20) $\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2}$

$= \frac{\left(\sqrt{\frac{I_1}{I_2}} + 1\right)^2}{\left(\sqrt{\frac{I_1}{I_2}} - 1\right)^2} = \frac{\left(\frac{13}{10} + 1\right)^2}{\left(\frac{13}{10} - 1\right)^2} = \left(\frac{23}{3}\right)^2$

21) Additional path difference introduction when the piston is moved through 8.75 cm = 2 × 8.75 = 17.50 cm

Since, the intensity changes from a maximum to a minimum, additional path = λ/2

∴ (λ/2) = 17.50 or λ = 35.0 cm

velocity, v = 350m/s = 35000 cm/s

 Frequency, $n = \frac{v}{\lambda} = \frac{35000}{35} = 1000Hz.$

22) (i) coefficient of t & x are of same sign so wave moves in –ve x direction.

(ii) wave speed = $\frac{5}{4}$ = 1.25 m/s

 In 2 sec wave covers = speed × time

= 1.25 × 2 = 2.5 m

(iii) $[y(x,t)]_{max} = \frac{0.8}{[4x + 5t]_{min^2} + 5} = \frac{0.8}{0 + 5} = 0.16m$

(iv) Value of y is never infinite, so it is travelling pulse.

23)

As direction of wave is reversed and wave will get inverted after reflection from rigid end or denser medium.

24) Let the frequency of first fork be n. Then the frequency of last fork in 2n. Since the successive tuning fork gives four beats, then
frequency of first fork = n
frequency of second fork = n + 4
frequency of third fork = n + 8 = n + 2 × 4
Frequency of Nth fork = n + (N – 1)4
As Nth fork is the last fork, so
2n = n + (N – 1)4
or n = (N – 1) 4 or 256 = (N – 1) 4
 N = 65

25)

$$V = \sqrt{\frac{\gamma RT}{M_W}}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{M_{W_2}}{M_{W_1}}} = \sqrt{\frac{32}{4}} = \sqrt{16} = 4:1$$

26) $x_1 \propto \frac{100+40}{2} - 40$

$x_2 \propto \frac{80+40}{2} - 40$

$\Rightarrow \frac{x_1}{x_2} = \frac{3}{2}$

27) (A) T = constant

(C) $WD_{on\ gas} = nRT \ln \frac{P_f}{P_i}$

(D) if T = const. ; PV = const.

28)

(i) Slope of line = $\frac{dT}{dt} = \frac{1}{ms} \propto \frac{1}{sp.\ heat}$

$(Slope)_1 > (Slope)_2$

so $\boxed{c_1 < c_2}$

(ii) Length of horizontal line ∝ Latent heat

so $\boxed{L_2 > L_1}$

29) (i) Work done by gas for path BC = 0

Work done by gas for path AB = $PV_0 = R(2T_0)$

(ii) Temperature at A = $2T_0$

Temperature at B = $4T_0$

(volume becomes double)

Temperature at C = $8T_0$

(pressure becomes double)

Heat absorbed = Heat absorbed A to B + Heat absorbed B to C

$= nC_P\Delta T + nC_V\Delta T$

$= \frac{5}{2}R(2T_0) + \frac{3}{2}R(4T_0) = 11RT_0$

(iii) $(nC_V\Delta T)_{AB} + (nC_V\Delta T)_{BC}$

$= \left[(1)\frac{3}{2}R(2T_0)\right] + (1)\frac{3}{2}R(4T_0)$

$= 9\ RT_0$

30) $k[100 - T_1] = 2k[T_1 - T_2] = k[T_2 - 20]$

$100 - T_1 = 2(T_1 - T_2) = T_2 - 20$

$100 - T_1 = 2T_1 - 2T_2$

$3T_1 - 2T_2 = 100$ .....(1)

$2T_1 - 3T_2 = -20$ .....(2)

on solving above equations, $T_1 = 68°C$, $T_2 = 52°C$

31) $\text{Ratio} = \frac{T \cdot K \cdot E}{R \cdot K \cdot E} = \frac{\frac{3}{2}kT}{\frac{2}{2}kT}$

$= \frac{3}{2}$

32) $\frac{dQ}{dt} = \sigma AT^4$ (for black body e = 1)

$\frac{1}{A} \times \left(\frac{dQ}{dt}\right) = \sigma T^4$

$R = \sigma T^4$

$T^4 = \frac{R}{\sigma}$

$T = \left(\frac{R}{\sigma}\right)^{1/4}$

Now, $\lambda_m T = b$

$\lambda_m = \frac{b}{T} = b\left(\frac{\sigma}{R}\right)^{1/4}$

33) $Q = \int_{T=1}^{T=2} mSdT = \left[\frac{mT^4}{5}\right]_1^2 \Rightarrow \frac{Q}{m} = \frac{15}{5} = 3$

34) For iso-thermal process, PV = constant

and for adiabatic process, PV^γ = constant (where $\gamma > 1$)

Process BC is isothermal & CA is adiabatic

$\Rightarrow P_0V_0 = 3P_0V_C$

$\Rightarrow V_C = \frac{V_0}{3}$ [for process BC]

$3P_0\left(\frac{V_0}{3}\right)^\gamma = \frac{V_0^\gamma P_0}{2}$ [Adiabatic]

$\Rightarrow \gamma = \frac{\ell n 6}{\ell n 3}$ [for process CA]

35) $P = \frac{Q}{t} = \frac{ms\Delta T}{t}$

$= 4200 \times \frac{20}{100} = \frac{2 \times 5 \times 4.2 \times \Delta T}{10}$

$\boxed{\Delta T = 200^{o}C}$

36) $T_1 = 273 + 27 = 300$ K

For constant volume $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

$\Rightarrow T_2 = T_1 \times \frac{P_2}{P_1} = 300 \times \frac{2}{1} = 600$ K

$= 600 - 273 = 327°C$

37) $PV = nRT$

$P(0.9V) = n'R(1.2T)$

$n' = \frac{3}{4}n$

Gas leaks out $= \frac{n}{4} = 25\%$

38)

$VT^2 = \text{constant}$

$\frac{\Delta V}{V} + \frac{2\Delta T}{T} = 0$

$\frac{\Delta V}{V} = \frac{-2\Delta T}{T}$

$\frac{\Delta V}{\Delta TV} = \frac{-2}{T}$

$\gamma = \frac{-2}{T}$

39) Ideal gas equation,

$PM_w = \rho RT$

Solution:

Now, analyzing each graph one by one.

(i) T = constant

$P \propto \rho$

as P(↑), ρ (↑)

(ii) P = constant

ρT = constant

as T(↑), ρ(↓)

(iii) $P \propto T$, ρ = constant

40)

From the given P–T diagram:

A → B: Vertical downward line ⇒ isothermal process, decrease in pressure

B → C: Straight line passing through origin (dashed extension) ⇒ constant volume (isochoric process)

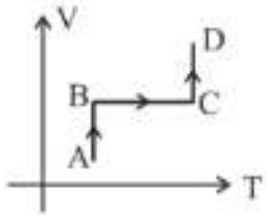
C → D: Vertical downward line ⇒ isothermal process, decrease in pressure.

Now convert to V–T diagram:

A → B (isothermal): T = constant, pressure decreases ⇒ volume increases → Vertical line upward in V–T graph

B → C (isochoric): Volume constant while temperature increases → Horizontal line to the right in V–T graph

C → D (isothermal): T = constant, pressure decreases ⇒ volume increases → Vertical line upward in V–T graph



41) $W = P\Delta V$

$W = nR\Delta T$ ....(1)

$Q = n C_p \Delta T$ ....(2)

(1) / (2)

$$\frac{W}{Q} = \frac{R}{C_p} = \frac{R}{\left(\frac{f+2}{2}\right)R} = \frac{2}{7}$$ (f = 5)

$$W = \frac{2}{7}Q$$

42) $P \propto T^4$

$$\frac{P}{P'} = \left(\frac{273+0}{546+273}\right)^4 = \left(\frac{273}{819}\right)^4 = \frac{1}{3^4}$$

$$\boxed{P' = 81P}$$

43)

Heat lost = Heat gain

$m S_A(32 - 28) = mS_B(28 - 24)$

$S_A = S_B$

44) $\lambda_m \propto \frac{1}{T}$
Since, $\lambda_{m_2} > \lambda_{m_1}$
$\Rightarrow T_2 < T_1$

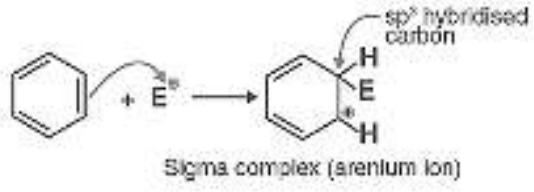
45) First resonating length = $\frac{\lambda}{4} = \frac{V}{n.4} = \frac{320}{320 \times 4}$
$= \frac{1}{4} m = 25cm$
next resonating length = $\frac{3\lambda}{4} = 75cm$
next resonating length = $\frac{5\lambda}{4} = 125cm$
next resonating length = $\frac{7\lambda}{4} = 175cm$ which is not possible
So max. length of air column at resonance = 125 cm
min. length of water column at resonance = 5 cm
min. length of air column at resonance = 25 cm
max. length of water column at resonance = 105 cm

CHEMISTRY

46) **Question Explanation:** The question asks to evaluate the validity of two statements regarding the σ-complex (Wheland intermediate) formed during the electrophilic substitution of benzene.

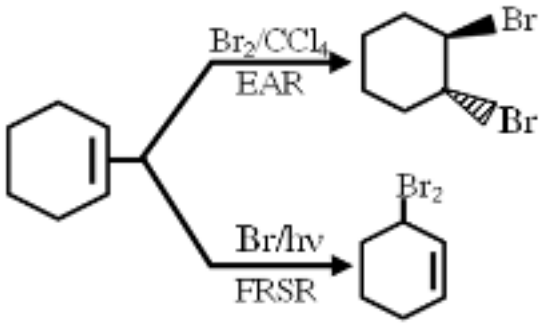
Concept: Electrophilic Aromatic Substitution, σ-complex, Hybridization, Aromaticity, Hückel's Rule.

Solution:
Statement-I: When an electrophile (E^+) attacks benzene, it forms a C–E bond using two electrons from the system. The carbon receiving the electrophile changes from sp^2 to sp^3 hybridization. (Correct)



Statement-II: The σ-complex is non-aromatic because the sp^3 carbon breaks the continuous overlap of p-orbitals, disrupting the cyclic resonance. (Incorrect)

Final Answer: Option (1)



47)

48)

This reaction is free radical substitution reaction.
Reactivity ∝ stability of radical.

49)

Correct Ans. (2)

50) NCERT-XI Part-II 23-24 P.No. 304

51)

A) Question Explanation: The question asks to correctly match methods/tests with their outcomes or purification techniques.

B) Concept Used: Chromatography, Lassaigne's test for N and S, Steam distillation, Vacuum distillation

C) Answer Explanation: Main Rule: Match each method with its characteristic result.

Column-I Item	Column-II Match	Reason
(A) Best and latest technique	(i) Chromatography	Modern separation method
(B) Lassaigne's test for N	(iv) Prussian blue coloured complex	NaCN + Fe → Prussian blue
(C) Lassaigne's test for S	(iii) Violet coloured complex	Na_2S + Nitroprusside → Violet
(D) Aniline purified from water	(v) Steam distillation	Aniline is steam volatile
(E) Glycerol purified from spent lye	(ii) Vacuum distillation	Glycerol decomposes at bp

The Correct matching order is (A)-(i), (B)-(iv), (C)-(iii), (D)-(v), (E)-(ii). This matches

Option 3.D)
Final Answer: Option (3)

52)

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53)

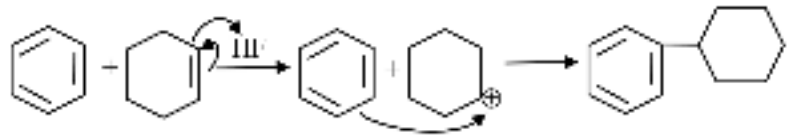
Correct Ans. (4)

54)

A) Question Explanation: The question asks to identify products 'A' and 'B' from two separate reactions starting with benzene.

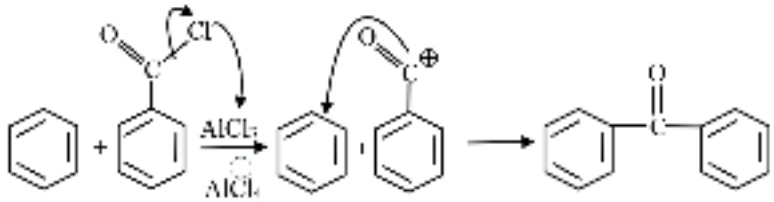
B) Concept Used: Friedel-Crafts Alkylation, Friedel-Crafts Acylation, Electrophilic Aromatic Substitution

C) Answer Explanation:
Reaction 1: Friedel-Crafts Alkylation



(A)

Reaction 2: Friedel-Crafts Acylation



(B)

D) Final Answer: The correct option is option 2 - A is cyclohexylbenzene and B is Benzophenone.

55)

Fact

56)

Correct Ans. (4)

57)

Correct Ans. (2)

58)

Correct Ans. (2)

59)

Correct Ans. (3)

60)

Correct Ans. (3)

61)

A → n-Hexune
B → Benzene
C → CH≡CH
D → Benzene

62)

Correct Ans. (4)

63)

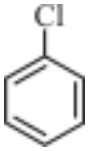
Correct Ans. (3)

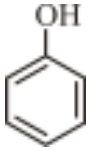
64) **Fact based.**
Addition of hydride on benzene ring to give carbocation.

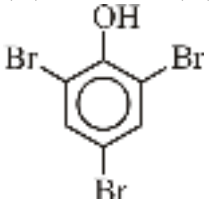
Correct answer is option (3).

65)

A = Hexane, B = Benzene

(C) 

(D) 



66)

While testing for halogens, lassaigne's extract is boiled with conc. HNO_3 to decompose Na_2S and NaCN if formed.

67)

A) Question Explanation: The question asks which type of compound can be separated by steam distillation.

B) Concept Used: Steam distillation, volatility, immiscibility with water.

C) Answer Explanation:

Steam distillation is used for compounds that are steam volatile but insoluble in water.

Such compounds co-distill with steam at lower temperatures, preventing decomposition.

Soluble or non-volatile compounds cannot be separated by this method.

D) Final Answer: Option (1) Steam volatile but insoluble in water

68)

Correct Ans. (2)

69)

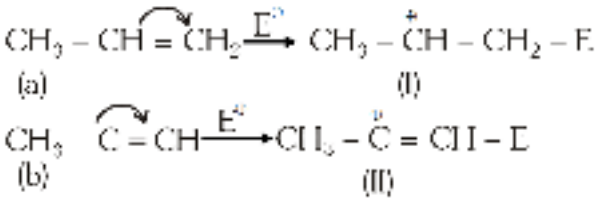
Correct Ans. (1)

70)

Correct Ans. (2)

71)

Correct Ans. (3)



72) (1)
I[st] carbocation is more stable than II
So (a) is more reactive than (b) towards EAR
(2) Only (b) reacts with Tollen's reagent

(3) $CH_3 - C \equiv CH \xrightarrow{Na/liq.NH_3} CH_3 - C \equiv \overset{-}{C}\overset{+}{Na}$

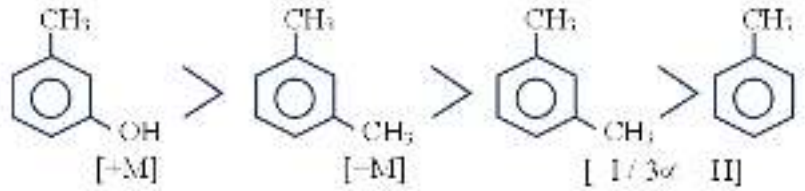
73)

Explaining the Question : Identify the correct order among the following

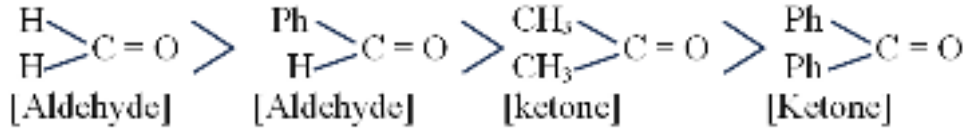
Concept :(1) Rate of ESR ∝ +I / +M group attached to benzene.

(2) Rate of NAR ∝ $\frac{1}{\text{stayc crowding around}}$ carbonyl carbon
[Aldehyde is more reactive then ketone]
(3) Rate of EAR ∝ stability of carbocation

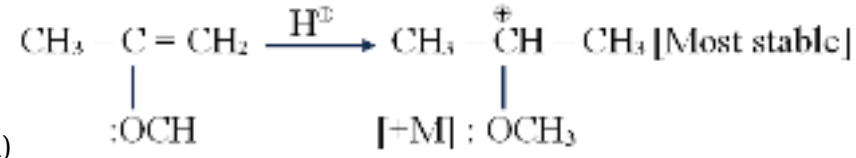
Solution :
(a)



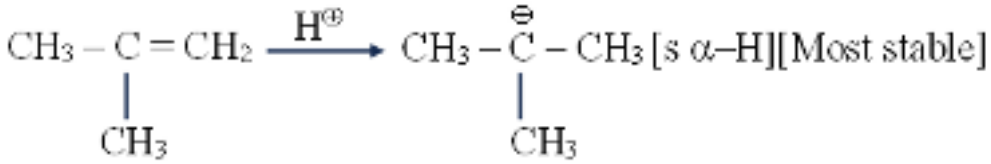
+M of –OH > $-CH_3$
Hence given order is correct
(b)



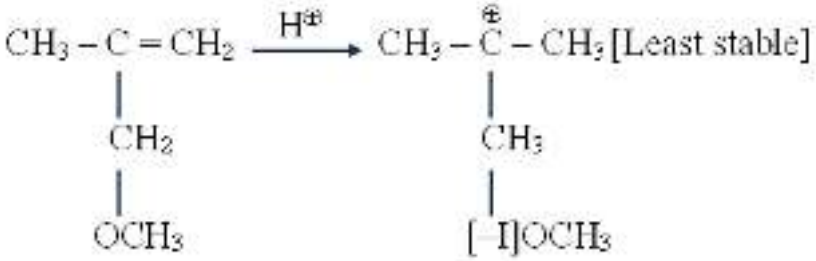
The given order is also correct.
(c)(i)



(ii)

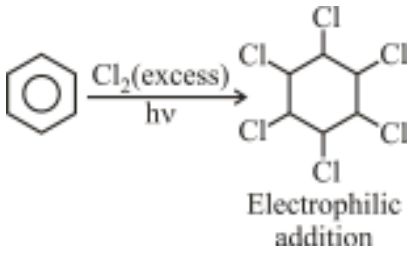


(iii)



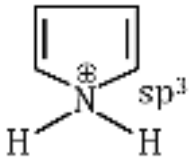
Hence order is (i) > (ii) > (iii) and is correct

Final answer :Option (1)

74) 

75)

Correct Ans. (2)

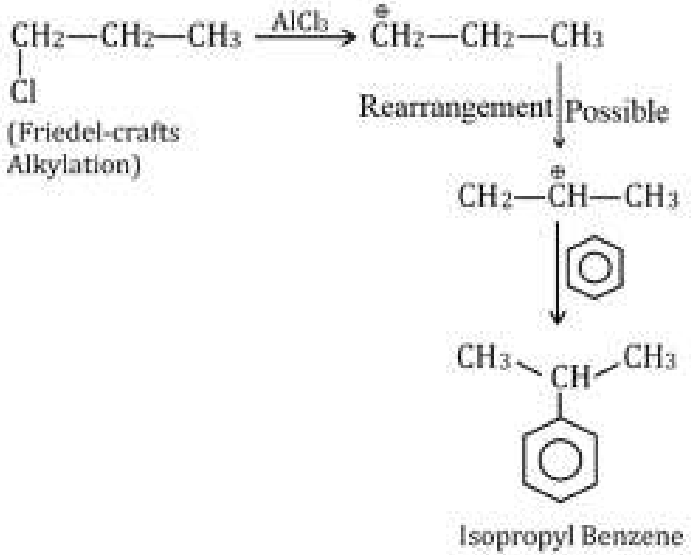
76) 

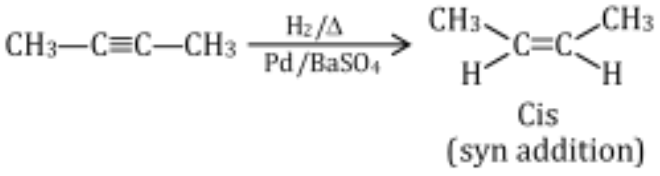
Non-aromatic

77) $CH_3 - COOH \xrightarrow[CaO]{NaOH} CH_4 + Na_2CO_3$

78)

Correct Ans. (1)

79) 

80) 

81)

Correct Ans. (4)

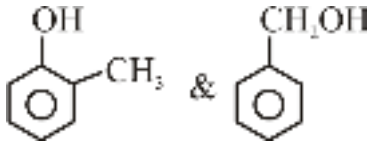
82)

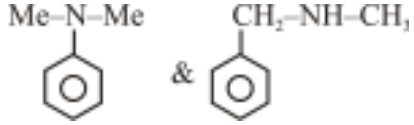
Correct Ans. (4)

83)

Correct Ans. (3)

84)

(i)  are F.G.I.

(ii)  are F.G.I

(iii) $CH_3–C\equiv C–CH_3$ &  are not Isomers.

C_4H_6 C_4H_8

(iv)  are chain Isomers

(v)  are not isomers.

85) Chloroform and aniline has sufficient difference in boiling point and therefore can be separated by distillation.

86)

Question Explanation:
The question asks the colour of precipitate formed when sodium fusion extract containing sulphur is treated with lead acetate.
Concept:
Lassaigne's test, sulphur detection, Pbs formation.
Solution:

A. Sodium fusion extract with sulphur gives Na_2S.
B. On treatment with lead acetate → $Pb^{2+} + S^{2-} \rightarrow PbS$.
C. Pbs is a black precipitate.

Final Answer: Option(3)

87)

Correct Ans. (3)

88) Correct Ans. (2)

89)

Correct Ans. (4)

90)

Correct Ans. (3)

BIOLOGY

91)

NCERT XI_ Pg. # 135

92) NCERT, Pg. # 136

93) NCERT, Pg # 137

94) NCERT, Pg. # 138

95) NCERT Pg. # 141

96) Module

97) NCERT Pg. # 141

98) NCERT, Pg. # 142, 143

99) NCERT, Pg. # 144, 146

100) NCERT Pg # 147

101) NCERT, Pg. # 149

102) NCERT XI, Pg. 145 + Module

103) NCERT-XI_Pg. # 147

104) NCERT Pg. # 139

105) NCERT Pg. # 134, 135

106) NCERT Pg. # 154

107)

NCERT XI_Pg. # 157

108) NCERT Pg. # 158

109) NCERT Pg. # 160

110) NCERT Pg. # 161-162

111) NCERT, Pg. # 157

112) NCERT Pg. No. # 155-156

113) NCERT Pg. # 158

114) NCERT Pg. # 159

115) NCERT Pg. # 160

116) NCERT, Pg. # 157

117) NCERT # 162

118)

NCERT Pg.#163-164

119)

NCERT XI Pg. # 163

120) NCERT XI Pg # 160

121) NCERT Pg. # 168

122) NCERT Pg. # 169

123) NCERT Pg. # 169

124)

NCERT-XI_Pg. # 174

125) NCERT, Pg. # 172

126) NCERT Pg. # 173

127) NCERT, Pg # 174

128) NCERT Pg. # 175

129)

NCERT Pg. # 176

130)

NCERT-XI_Pg.# 176

131) NCERT Pg # 177

132)

NCERT-XI_Pg. # 177

133) NCERT, Pg. # 178

134) NCERT Pg. 173

135)

NCERT-XI_Pg.# 173

136) NCERT Page No. # 320, 321

137) NCERT XI, Pg. # 319

138)

NCERT XI Pg. # 233

139)

NCERT PAGE NO. 232

140) NCERT-Pg#316

141) NCERT XI Pg. No. # 232

142) NCERT (XII) ; Page No. # 231

143) NCERT Pg.#236.

144) NCERT Pg. No. # 232, 233, 234

145) NCERT Pg. 232

146) NCERT (XII) ; Page No. # 317 (E)
NCERT (XII) ; Page No. # 317 (H)

147) NCERT Pg. # 233

148) NCERT Page # 335(E), 336(H)

149) NCERT Page # 338

150)

NCERT XI Pg. No. # 245

151)

NCERT XI Pg. No. # 247

152) NCERT, Pg # 334-335

153)

NCERT Pg. # 333

154)

NCERT XI Pg. No.# 241

155)

NCERT XI Pg. No. # 241-244

156)

NCERT 11th Pg. No. 336

157) NCERT Pg.# 331, fig 22.1

158) NCERT Pg. # 240, 241

159)

NCERT XII Page No. # 332

160) NCERT Page no. 336, 337

161) Module No. # 7, Pg. No. # 73, 74, 91

162)

NCERT Page # 335

163)

NCERT XI Page # 334 (II Para)

164)

NCERT Page # 333

165)

NCERT XI Pg. No. # 243

166) NCERT Pg.# 332

167)

NCERT XI Pg. No. # 226

168) NCERT-XI, Page No. 307

169) NCERT Pg. # 227

170)

NCERT XI Pg. No. # 227

171) NCERT XI Pg. No. # 219

172)

NCERT XI Pg. No. # 226

173)

NCERT XI Pg. No. # 221

174)

NCERT XI Pg. No. # 219

175)

NCERT XI Pg. No. # 227

176)

NCERT XI Pg. No. # 227

177) NCERT Pg. # 311

178)

NCERT XI Pg. No. # 225

179)

NCERT XI Pg. No. # 227

180)

NCERT-XI, Pg. # 309